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5.1 INTRODUCTION

5.1.1 This chapter has been prepared by Brookbanks Consultants Ltd and sets out the results of an assessment of the traffic-related environmental effects of the Proposed Development.

5.1.2 Data used in the assessment has been drawn from the revised Transport Assessment (TA) for the Proposed Development. The TA is attached as an appendix to the ES further information submission, found in **Appendix 7a**. The TA sets out transport issues relating to the Proposed Development and identifies any necessary interventions to mitigate the anticipated transport effects and to improve accessibility and safety for all modes of travel. This revised ES chapter should be read in conjunction with the TA.

5.1.3 Impacts of the development have not changed substantially from the previous assessment, submitted in support of the application. However, mitigation measures have been revised for the two junctions at Gablecross Roundabout (conversion to a signalised junction) and Merlin Way / A419 slip road roundabout (addition of traffic signals on approach arms). This is as a result of traffic flows generated by the new development in addition to anticipated traffic growth for the Swindon Borough.

5.1.4 This revised ES chapter responds to comments raised by Swindon Borough Council (SBC), JMP Consultants and Highways England. All revisions follow collaborative dialogues and workshops with the aforementioned parties that have agreed the scope of the revised material which includes changes in land use, updated planning policies, updated traffic flows and all other relevant updates throughout the project.

5.2 ASSESSMENT METHODOLOGY

Scope

5.2.1 For the purposes of this assessment, the Study Area corresponds to that used in the Transport Assessment. This is defined by the geographical extent of the traffic model being used to assess the highway impacts and primarily includes the highway corridors of the A419 and the A420.

Data sources

5.2.2 Detailed discussions have been held with Swindon Borough Council (SBC) and the Highways Agency (HA) to agree the necessary assessment methodology. This resulted in the use of the Saturn and Paramics traffic models which provided the traffic flows on which the assessment has been based.

Assessment approach

5.2.3 The methodology follows current best practice by assessing the impacts on the hierarchy of transport modes: pedestrians; cyclists; public transport users; and vehicle drivers and passengers.

5.2.4 The approach adopted for the traffic and transport assessment has been based on the Institute of Environmental Assessment (now IEMA) *Environmental Assessment of Road Traffic* (1993) (IEMA Guidance), which recommends screening criteria of:

1. Roads where traffic flow would increase by more than 30% as a consequence of a proposed development; or
2. Roads where traffic flows would increase by 10% and pass close to or through sensitive areas.

5.2.5 For the purposes of this assessment, the majority of the routes in the vicinity of the Site are considered to be sensitive as there are residential properties fronting onto the carriageways and there is also the potential for high volumes of pedestrian and cycle movements within the urban area.

5.2.6 The significance of each effect has been considered against the criteria within the IEMA Guidance, where possible. However, this state that:-

> "…for many effects there are no simple rules or formulae which define the thresholds of significance and there is, therefore, a need for interpretation and judgement on the part of the assessor, backed-up by data or quantified information wherever possible. Such judgements will include the assessment of the numbers of people experiencing a change in environmental impact as well as the assessment of the damage to various natural resources. "

5.2.7 The degree of each potentially significant effect has also been considered and an assessment has been made as to whether the Proposed Development would result in minor, moderate or major adverse impacts or would be beneficial. The criteria used to determine the significance of each of the traffic-related environmental effects is based on the advice given in the IEMA Guidance as summarised below.

Severance

5.2.8 Severance is the perceived division that can occur within a residential area if it becomes separated by a major traffic artery and is used to describe the factors that separate people from other people and places. For example, severance may be affected as a result of an increase in traffic that could affect the difficulty of crossing a road.

5.2.9 The effects of severance can be applied to motorists, pedestrians or residents. The IEMA Guidance suggests that changes in traffic flow of 30%, 60% and 90% are regarded as producing "minor", "moderate" and "major" changes in severance respectively. However, there are no predictive formulae which give simple relationships between traffic factors and levels of severance. The IEMA Guidance states that marginal changes in traffic flow are unlikely to create or remove severance.

Driver delay

5.2.10 Delays to existing traffic can occur at several locations within the local highway network as a result of the additional traffic that would be generated by a development. The IEMA Guidance states that delays are only likely to be significant when the traffic on the network surrounding the development is already at, or close to, the capacity of the system.

5.2.11 The theoretical capacity of a particular junction can be determined by assessing the Ratio of Flow Capacity (RFC) for priority controlled junctions and Degree of Saturation for signal controlled junctions. When an RFC value of 0.85 or more is experienced, or a degree of saturation of 90%, queuing and congestion are likely to occur during busy periods.

Pedestrian delay

5.2.12 Changes in the volume, composition or speed of traffic may affect the ability of people to cross roads, and therefore increases in traffic levels are likely to lead to greater increases in delay. Delays are dependent upon the general level of pedestrian activity and general physical conditions of the crossing location.

5.2.13 Given the range of local factors and conditions which can influence pedestrian delay, the IEMA Guidance does not recommend that thresholds be used as a means to establish the significance of pedestrian delay, but recommends that reasoned judgements be made instead. However the IEMA Guidance does note that, when existing traffic flows are low, increases in traffic of around 30% can double the delay experienced by pedestrians attempting to cross a road.

Pedestrian amenity

5.2.14 Pedestrian amenity is broadly defined as the relative pleasantness of a journey, and is considered to be affected by traffic flow, traffic composition and pavement width/separation from traffic.

5.2.15 The IEMA Guidance notes that changes in pedestrian amenity may be considered significant where the traffic flow is halved or doubled, with the former leading to a beneficial effect and the latter to an adverse effect.

Fear and intimidation

5.2.16 The scale of fear and intimidation experienced by pedestrians is dependent on the volume of traffic, its HGV composition, its proximity to people or the lack of protection caused by such factors as narrow pavement widths, as well as factors such as the speed and size of vehicles.

5.2.17 There are no commonly agreed thresholds by which to determine the significance of this effect. However, the IEMA Guidance notes previous work that has been undertaken which puts forward thresholds that define the degree of hazard to pedestrians by average traffic flow, 18 hour/day heavy vehicle flow and average speed over an 18 hour day in miles per hour.

5.2.18 The IEMA Guidance also notes that special consideration should be given to areas where there are likely to be particular problems, such as high speed sections of road, locations of turning points and accesses. Consideration should also be given to areas frequented by school children, the elderly and other vulnerable groups.

Accidents and safety

5.2.19 Where a proposed development is expected to produce a change in the character of the traffic on the local road network, as a result of increased HGV movements for example, the IEMA Guidance states that the implications of local circumstances or factors which may elevate or lessen risks of accidents, such as junction conflicts, would require assessment in order to determine the potential significance of accident risk.

Significance criteria

5.2.4 The table below sets out the criteria for determining the magnitude of changes, the sensitivity of receptors and the significance of effects, with these being based on the IEMA Guidance.

Table 5.1: Magnitude

Magnitude	Criteria
Large	Changes in total traffic or HGV flows over 90%
Moderate	Changes in total traffic or HGV flows of 60% - 90%
Small	Changes in total traffic or HGV flows of 30% - 60%
Negligible	Changes in total traffic or HGV flows less than 30%

Table 5.2: Sensitivity

Sensitivity	Criteria
High	where the Proposed Development could be expected to have a very significant environmental effect (either adverse or beneficial) on severance, driver stress and delay, pedestrian and cyclist amenity, fear and intimidation, and accidents and safety during the construction and operational phases
Medium	where the Proposed Development could be expected to have a noticeable environmental effect (either adverse or beneficial) on severance, driver stress and delay, pedestrian and cyclist amenity, fear and intimidation, and accidents and safety during the construction and operational phases
Low	where the Proposed Development could be expected to result in a small, barely noticeable environmental effect (either adverse or beneficial) on severance, driver stress and delay, pedestrian and cyclist amenity, fear and intimidation, and accidents and safety during the construction and operational phases
Negligible	where no discernible environmental effect is expected (less than 30%, or 10% in sensitive areas) as a result of the Proposed Development on severance, driver stress and delay, pedestrian and cyclist amenity, fear and intimidation, and accidents and safety during the construction and operational phases

Table 5.3: Significance

MAGNITUDE	SENSITIVITY			
	High	**Medium**	**Low**	**Negligible**
Large	Major	Major	Moderate	Minor
Moderate	Major	Moderate	Minor	Negligible
Small	Moderate	Minor	Minor	Negligible
Negligible	Minor	Negligible	Negligible	Negligible

5.3 RELEVANT POLICY

National Planning Policy Framework (March 2012)

5.3.1 Paragraph 30 within the NPPF states that encouragement should be given to solutions which support reductions in greenhouse gas emissions and reduce congestion. Local planning authorities should therefore support a pattern of development which, where reasonable to do so, facilitates the use of sustainable modes of transport.

5.3.2 Paragraph 32 states that all developments that generate significant amounts of movement should be supported by a Transport Statement or Transport Assessment. Plans and decisions should take account of whether:

- The opportunities for sustainable transport modes have been taken up depending on the nature and location of the site, to reduce the need for major transport infrastructure;
- Safe and suitable access to the site can be achieved for all people; and
- Improvements can be undertaken within the transport network that cost effectively limit the significant impacts of the development. Development should only be prevented or refused on transport grounds where the residual cumulative impacts of development are severe.

5.3.3 Paragraph 35 states that plans should protect and exploit opportunities for the use of sustainable transport modes for the movement of goods or people. Therefore developments should be designed where practical to:

- Accommodate the efficient delivery of goods and supplies;
- Give priority to pedestrian and cycle movements and have access to high quality public transport facilities;
- Create safe and secure layouts which minimise conflicts between traffic and cyclists or pedestrians, avoiding street clutter and where appropriate establishing home zones; and
- Consider the needs of people with disabilities by all modes of transport.

5.3.4 A key tool will be a Travel Plan, as identified in paragraph 36. All developments which generate significant amounts of movement are required to provide a Travel Plan.

5.3.5 Paragraph 37 states that planning policies should aim for a balance of land uses that minimise journey lengths for employment, shopping, leisure, education and other activities. Developments should promote a mix of uses in order to provide opportunities to undertake day-to-day activities including work on site.

5.3.6 When setting local parking standards for residential and non-residential development, paragraph 39 states that local planning authorities should take into account:

- Accessibility of the development;
- The type, mix and use of the development;
- The availability of and opportunities for public transport;
- Local car ownership levels; and
- An overall need to reduce the use of high-emission vehicles.

National transport guidance

5.3.7 **Manual for Streets 1 and 2 (MfS):** The UK Department for Transport (DfT) and the Department for Communities and Local Government (DCLG), with support from the Commission for Architecture and the Built Environment (CABE), commissioned WSP Group, Transport Research Laboratory (TRL), Llewelyn Davies Yeang and Phil Jones Associates to develop Manual for Streets to give guidance to a range of practitioners on effective street design.

5.3.8 **The Manual for Streets (March 2007)** guidance on the planning, design, provision and approval of new streets, and modifications to existing ones. It aims to increase quality of life through good design which creates more people-oriented streets. The detailed guidance applies mainly to residential streets although the overall design principles can be applied to all streets within urban areas.

5.3.9 A street is defined as "a highway with important public realm functions beyond the movement of motor traffic" – i.e. by its function rather than just the road hierarchy.

5.3.10 Manual for Streets has updated geometric guidelines for low trafficked residential streets, examines the effect of the environment on road user behaviour, and draws on practice in other countries. This research provides the evidence base upon which the revised geometric guidelines in the Manual for Streets are based, including link widths, forward visibility, visibility splays and junction spacing.

5.3.11 Manual for Streets 2 - Wider Application of the Principles is the result of collaborative working between the Department for Transport and the transportation industry.

5.3.12 The aim of the document is to extend the advantages of good design to streets and roads outside residential areas, largely covered in MfS1. By amending the way high streets and non-trunk roads are designed, the fabric of public spaces and the way people behave can be changed. It means embracing a new approach to design and breaking away from inflexible standards and traditional engineering solutions.

5.3.13 The new guide does not supersede Manual for Streets 1, rather it explains how the principles of the first document can be applied more widely.

5.3.14 **Design Manual for Roads & Bridges:** The DfT publish a large suite of documents known as the Design Manual for Roads and Bridges, which provides detailed standards and guidance on the provision of highway networks. The suite of documents provides a comprehensive manual which accommodates all current standards, advice notes and other published documents relating to the design, assessment and operation of trunk roads including motorways. The standards are routinely adopted by local highway authorities for their local highway network.

5.3.15 **HE Circular 02/2013:** The Strategic Road Network and the Delivery of Sustainable Development: The Circular was published in 2013 and explains how the HE will engage with the planning system and provides details on how the HE will fulfil its remit to be a delivery partner for sustainable economic growth whilst maintaining, managing and operating a safe and efficient strategic road network.

5.3.16 The Circular identifies that development proposals are likely to be acceptable if they can be accommodated within the available highway capacity on the strategic road network, or they do not increase demand for use of a section that is already

operating at over-capacity levels, taking account of any travel plan, traffic management and/or capacity enhancement measures that may be agreed. Furthermore it is noted that Paragraph 9 identifies that development should only be prevented or refused on transport grounds where the residual cumulative impacts of development are severe.

5.3.17 Paragraph 25 identifies that the overall forecast demand should be compared to the ability of the existing network to accommodate traffic over a period up to ten years after the date of registration of a planning application.

5.3.18 Paragraph 27 identifies that where the overall forecast demand at the time of opening of the development can be accommodated by the existing infrastructure, further capacity mitigation will not be sought.

5.3.19 With regard to travel plans Paragraph 29 and 30 highlights that it may be possible to free up additional capacity within the road network so that the demand generated by a proposed new development, which would otherwise be unacceptable, can be accommodated.

5.3.20 Paragraph 34 identifies that at locations where there insufficient capacity, the impact of the development will be mitigated to ensure that the strategic road network is able to accommodate existing and development generated traffic.

5.3.21 In relation to providing new access points, Paragraph 39 identifies that where appropriate, proposals for the creation of new junctions or direct means of access may be identified and developed at the Plan-making stage in circumstances where it can be established that such new infrastructure is essential for the delivery of strategic planned growth

Swindon Borough Local Plan 2026

5.3.22 The Swindon Borough Local Plan 2026 is the main planning policy document for Swindon Borough, providing the planning policy framework to deliver sustainable growth up to the year 2026. The Local Plan identifies where, when, and how development will take place in Swindon. The local plan was formally adopted by Swindon Borough Council on 26th March 2015.

5.3.23 The Local Plan presents an overall spatial Vision for Swindon as indicated below:

> "The Borough of Swindon will become a place where people choose to live, visit and invest. As an important regional centre, Swindon's appeal will stem from having an attractive and well-equipped town that has successfully blended traditional architecture with high quality contemporary buildings that incorporate sustainable design and construction principles."

5.3.24 To achieve the overall Vision, 10 strategic objectives have been identified. Those relating to transport are highlighted below:

Strategic Objective 1: High Quality Sustainable Development – to improve the image of Swindon, enable inclusive communities and address climate change by the provision of high quality, well designed and sustainable development.

Strategic Objective 2: Infrastructure - to meet the infrastructure needs for and arising from the growth of Swindon (including health and community needs) in a timely and co-ordinated manner and being adequately funded.

Strategic Objective 7: Transport - to support Swindon's growth through the provision of a comprehensive and sustainable transport network that is efficient, safe, affordable, accessible and easy to understand, and offers a genuine choice of modes.

Strategic Objective 9: Green Infrastructure - to provide an attractive and inspirational environment to live, work, learn and play, by the provision of a far-reaching network of connected and multi-functional green spaces linked to the wider countryside.

5.3.25 To support the Strategic Objectives, various policies have been proposed.

Policy SD1: Sustainable Development Principles - To enable the delivery of sustainable development and support sustainable communities in the Borough.

Policy SD2: The Sustainable Development Strategy

Recognising its role and function in the wider area development in the Borough will be concentrated at Swindon through a combination of realising development opportunities within Swindon's urban area and allocated strategic sites at Wichelstowe, Commonhead, Tadpole Farm, Kingsdown and the proposed New Eastern Villages, Rowborough and expanded South Marston.

Policy SD3: Managing Development. When considering development proposals, the Council will take a positive approach to reflect a presumption in favour of sustainable development.

5.3.26 To support the delivery of the policies, several overall 'Themes' are outlined. Theme 4 relates to Transport. This identifies specific policies to:

- Improve key transport gateways and corridors
- Provide good access to Swindon Town Centre and key destinations
- Reduce severance caused by transport corridors and the dominance of the car on the street scene
- Minimise the environmental impact from transport (for example, vehicle emissions)
- Minimise congestion and therefore journey time, noise and air quality
- Promote healthy lifestyles and travel choices and maximise opportunities to walk and cycle
- Support good public transport provision
- Encourage innovative transport initiatives for rural areas

Policy TR1: Sustainable Transport Networks - The council will use its planning and transport powers to help reduce the need to travel, and support and encourage the sustainable, safe and efficient movement of people and goods within and through the Borough.

Policy TR2: Transport and Development - New development should be located and designed to reduce the need to travel and to encourage the use of sustainable transport alternatives, particularly walking and cycling, and provide the potential to maximise bus travel. Development shall be permitted where proposals provide access that is appropriate to the scale, type and location without detriment to highways safety and local amenity, and where there is an existing safe and

convenient pedestrian and cycle access or provision is made for such access. Development shall be permitted where proposals will not prejudice or impede an existing or planned cycle route, or provision is made for a more satisfactory route. Development that results in the loss of existing rights of way or their disruption shall only be permitted when adequate, acceptable alternative provision or diversions be arranged. Development shall be permitted where proposals do not remove, narrow or materially impair the approved line of the Thames Path or Ridgeway National Trails, and/or public access to them. Developments shall provide appropriate mitigating measures to offset any adverse impacts on the transport network at both the construction and operational stages. Parking provision, including secure cycle and motorcycle parking, should be provided in accordance with the Council's adopted parking standards.

Policy NC3: New Eastern Villages - including Rowborough and South Marston Village Expansion

Land to the East of the A419, as defined on the Proposals Map, is allocated for a mixed-use development. The form of the development shall comprise a series of new distinct villages and an expanded South Marston village. The development shall provide:

- about 6,000 dwellings at the New Eastern Villages (south of the A420)
- about 1,500 dwellings at Rowborough (north of the A420)
- 500 dwellings at South Marston.
- Affordable housing at Rowborough and South of the A420 in accordance with Policy HA2. Separate proposals are made in Policy RA3 in relation to South Marston.
- High quality public realm including outdoor civic public space;
- sustainable transport links including a a rapid bus transit link through the district centre that connects the Eastern Villages to Swindon Town Centre as part of phase 1 of the development which includes residential development north of the A420, the District Centre and the employment allocation;
- Walking and cycling network improvements that integrate with existing networks and provide good connectivity to the surrounding area;
- an improved gateway junction at White Hart to manage additional demand and deliver high quality public realm;

- improvements to the Oxford Road/Drakes Way and Covingham Road/Dorcan Way transport corridors;
- a green bridge across the A419 near Covingham Drive to provide for walking, cycling and public transport;
- a new road link to the Commonhead Roundabout;
- a new road link under or across the Bristol to London railway line connecting the development north and south, and new and/or improved accesses to the A420 for proposed residential and employment uses;
- a 1000 (3ha.) space Park and Ride site; and
- measures to minimise rat-running through existing adjacent villages and east Swindon.
- An An extensive green infrastructure network that maximizes opportunities for habitat connectivity and enhanced biodiversity including extending the River Cole green infrastructure corridor and connecting with Nightingale Wood
- Sports and leisure facilities, including playing pitches a leisure center and a 25m swimming pool;
- about 40 hectares of employment land in total (B Class Uses) to be located south and east of the A420 and A419 respectively adjacent to the White Hart Junction, in a form that complements the mixed-use District Centre, and safeguards its connection with the wider residential development in the Eastern Villages and distributed as follows:
    - 2.5 hectares within B1a use-class to be located at and/or adjacent to the District Centre;
    - 7.5 hectares within B1b/c or B2 use-class; and
    - 30 hectares within B8 use class.
- about 12,000m2 (gross) of retail floor space including a high quality District Centre with strong connectivity to the adjacent residential areas, comprising an anchor food store and complementary uses, and a network of Local Centres that offer retail provision of a scale that meets the daily shopping needs of the communities that they serve, including the existing community of South Marston;
- Educational educational requirements comprising:
- a minimum of 8 -forms of entry of secondary provision;
- a minimum of 8 forms of entry of primary provision with early learning facilities;
- a site for a special school for children and young persons with profound, multiple and severe learning difficulties for ages 0-25, and

- measures to manage the demographic peak at primary and secondary schools;
- Community facilities by the means of safeguarded land and/or developer contributions, including where appropriate flexible, multi-purpose buildings for use by the community, the public sector and for worship at the district centre, local centres and the learning campus;
- A health care facility with GP, dentist and pharmacy at the District Centre
- Safeguarded land for a fire station towards the southern part of the site, and
- A sewage treatment works if required.

5.3.27 Policy Policy RA3 in respect of South Marston is also relevant. It addresses the expansion of the village. It makes provision for respecting the character of the village, for affordable housing, for education, community and recreation facilities at the heart of the village, for green infrastructure. In respect of transport it provides for a new road connection between Thornhill Road and Old Vicarage Lane seeking that the connection should be designed (within the expanded village) so that it does not form a bypass to South Marston. It provides for *"traffic management and sustainable transport measures to minimize the volume of traffic, including traffic passing through the village, and to reduce the impact on the village, in particular at Pound Corner)"*

Infrastructure Delivery Plan

5.3.28 SBC is required to demonstrate that the policies in the Local Plan will be delivered in a sustainable way. To this end, there is increased emphasis on the need to identify infrastructure required to support future growth, resulting from additional housing and employment during the Plan period to 2026. The IDP focuses on those infrastructure items which will require capital expenditure.

5.3.29 The IDP helps to identify:

- Types and costs of infrastructure
- Delivery timetable
- Gaps in funding
- Critical pieces of local and strategic infrastructure.

5.3.30 In relation to transport infrastructure, the WSP *Swindon Transport Strategy* (2009) sets out Swindon's transport aspirations to 2026.

Rapid Transit

5.3.31 The Northern Rapid Transit link will deliver the first arm of Swindon's Rapid Transit scheme connecting Tadpole Farm and the Northern Development Area with Swindon town centre. The rapid transit routes from Wichelstowe and the Eastern Villages will provide fast and reliable journey times for new communities to Swindon Town Centre and complete the rapid transit network. The rapid transit network will complement the Park and Ride Strategy – additional sites are proposed at Wichelstowe, Eastern Villages and Commonhead.

Town Centre

5.3.32 The new Town Centre bus exchange and improved bus stops will support economic growth by providing a viable and attractive alternative to the private car for everyday journeys.

Eastern Villages including Rowborough and South Marston

5.3.33 The Eastern Villages will deliver a new district centre, not just to serve the new community but also the existing residential areas at Covingham, Dorcan and Parks. The main route from the New Eastern Villages into Swindon's Central Area will be via Oxford Road (A420) and White Hart Roundabout. The intention is to deliver an improved junction at White Hart that establishes strong linkages between the Eastern Villages and the rest of Swindon, prioritises access into Swindon Town Centre and improves the public realm. A Green Bridge is proposed that will provide access across the A419 – part of the Strategic Road network and a significant physical barrier for public transport, walking and cycling. The Commonhead Link (and associated Junction 15 improvements) will provide direct access to the Strategic Road Network (A419 and M4). Junction improvements will also be required on the A420.

Draft Supplementary Planning Document 2009 – Eastern Development Area

5.3.34 The SPD has since been redrafted in light of representations made, changes to the emerging Swindon Core Strategy, and to reflect new evidence based studies that were commissioned following the first draft.

5.3.35 The 2009 SPD identified that the development area could accommodate a potential 12,000 dwellings together with a range of supporting land uses. The purpose of the document is to provide a planning framework setting out the expectations and requirements of development for the EDA.

5.3.36 The draft SPD identified a set of development principles, as detailed below:

- Deliver a diverse living environment of high quality that accommodates growth targets within an efficient and generally compact urban form with a strong sense of place.
- Create improved eastern approach routes into Swindon and help redefine the image of Swindon.
- Form part of Swindon and integrate as seamlessly as possible with the eastern edge of town.
- Create new sustainable communities and facilities that connect with and protect the character of existing surrounding communities.
- Support the broader Swindon economy with new employment and investment opportunities.
- Become part of and help deliver a new improved strategic transport network for Swindon that encourages public transport, pedestrian and cycle use, seeking to achieve connectivity within and beyond the EDA.
- Assist the physical, social and economic regeneration of East Swindon.
- Conserve and enhance natural systems, watercourses, biodiversity and landscape resources, as part of an integrated green infrastructure proposal.
- Conserve and enhance Swindon's cultural heritage, recognising its key assets and preserving the identity of those assets within the EDA.
- Connect Swindon with its wider landscape setting in a sympathetic way.

5.3.37 The draft SPD identified three potential broad layouts for the development, as identified below:

The Crunch: The main premise of 'The Crunch' is that the EDA develops as an integral part of Swindon that, as seamlessly as possible, joins new development to east Swindon. 'The Crunch' seeks to overcome severance caused by the A419,

which will otherwise be a barrier to integration between the EDA and the existing eastern edge of town.

The Stretch: In this option it is accepted that the major barriers to integration with Swindon's existing urban area are too difficult or expensive to overcome and that the EDA must establish its own distinctive and separate identity. 'The Stretch' effectively turns the EDA into a Separate community on the edge of Swindon. Whilst it may not directly turn its back to the town, it is not driven by a fundamental aim to integrate with the urban area. This means that the physical centre of the EDA is located in the centre of the new development area – stretched away from east Swindon thereby reinforcing its separate identity.

The Weave: This is a hybrid option that shares the basic philosophy of the crunch but recognises that some of the physical severance issues are challenging if they are to be overcome, at least in the short to medium term. 'The Weave' sees the EDA as an integrated part of Swindon but takes a stepped approach to delivering it. One of the important features of 'The Weave' is that it could evolve into 'The Crunch' at some later stage.

5.3.38 The draft SPD identified that the Crunch was at the time of writing preferred.

5.3.39 Subsequent to the production of the draft SPD, additional background studies have been carried out. This included SBC commissioning JMP to undertake some initial traffic modelling. This work utilised the A419 Paramics model. The study focused on the potential transport impact of growth to the east of Swindon on the A419, M4 J15 and J16 and on the A420 east of the White Hart Junction. This report considered 7,500 homes and associated services and facilities up to 2026, which corresponds with the timeframe of the Core Strategy. An updated Halcrow SATURN Model has been used as the basis for this PARAMICS assessment.

5.3.40 This additional study identified that the White Hart junction in its current form would not have sufficient capacity to cater for the anticipated increases in traffic volumes arising from the Eastern Villages even if mitigation is provided elsewhere such as providing a full or partial all vehicular link at the Green Bridge. The report then

suggested that the existing roundabout be amended to reflect a diamond interchange.

5.3.41 A basic comparison of journey times that the operation of a part signalised White Hart roundabout with modified slips would operate just as well / slightly better than the diamond interchange in terms of journey times.

5.3.42 The report identified that the extent of works required to the A420 is largely dictated by the need to upgrade existing junctions and provide new junctions along the corridor. These junctions generally require two or more approach lanes in each direction. The western site access and Gablecross are situated sufficiently close together as to necessitate the widening of the remainder of the A420 to a two lane standard as a minimum in the westbound direction.

5.3.43 In total, three existing junctions would need to be upgraded (Gablecross, Police Station and Old Vicarage Lane) and two new junctions constructed in order to provide a western and eastern access into the development.

5.3.44 The existing junctions would need to be upgraded so as to provide a balance between demand for A420 through movements and that from the Eastern Villages and South Marston. At Gablecross the replacement of the existing roundabout with a signalised crossroads would provide additional capacity and would facilitate pedestrian movements.

5.3.45 The report identified that the Commonhead to M4 J15 improvements associated with the Commonhead planning application should be sufficient to accommodate the Eastern Villages if the focus of the development is along the A420.

Eastern Villages Draft Supplementary Planning Document 2013 :

5.3.46 The draft SPD replaced the 2009 Draft SPD for the Eastern Development Area. While it is no longer being progressed beyond the draft stage, it is still referred to in the adopted Local Plan as a key evidence base document. It identified a set of development principles, as detailed below:

- Sets out the overall vision and principles

- Seeks to ensure that the Vision for the development is delivered in accordance with the key principles
- Provides a detailed planning framework
- Provides a catalyst to express aspirations for the development
- Offers detail and clarity on the expected outcomes from the development

5.3.47 The SPD identifies the overall Vision for the Eastern Village Development. The Eastern Villages development will create sustainable well-designed places where people want to live, work and spend time. The development will:

- Be a high quality, sustainable development with strategic infrastructure that benefits Swindon as a whole
- Sensitively and positively respond to the existing landscape context, natural and historic assets and the character and identity of the surrounding villages as well as enhancing biodiversity and green infrastructure;
- Comprise new distinct villages with individual identities and characters linked together by green spaces that integrate with the existing urban area and wider landscape setting;
- Provide the facilities and services required for the new communities and opportunities for existing communities to benefit from the development
- Form a new eastern gateway to Swindon that improves the image of the Town and maximises benefits to the wider economy including connections to the Town Centre.

5.3.48 The key principles will be to:

- Create vibrant, attractive places that are functional, durable and capable of adapting to accommodate changing lifestyles in line with sustainable communities
- Ensure the timely delivery of necessary facilities to support these sustainable communities
- Conserve and enhance natural systems, watercourses, biodiversity and landscape settings, including integration with green infrastructure to help mitigate and facilitate adaptation to climate change
- Conserve and enhance Swindon's historic assets and cultural heritage, allowing their use and enjoyment without harming them for future generations

- Ensure the development becomes part of the wider town and supports the physical, social and economic regeneration of East Swindon. Respect and support the existing character and setting of surrounding villages;
- Contribute to and not harm the Town Centre regeneration and support the broader Swindon economy with new employment and investment opportunities;
- Contribute to an improved strategic transport network for Swindon and facilitate public transport priority for journeys into the Town Centre;
- Encourage walking and cycling, within the development and ensure high levels of accessibility and connectivity within and beyond the new Eastern Villages
- Enhance the image of Swindon by maximising place-making opportunities particularly around key nodes, gateways and frontages.

5.3.49 The SPD identifies that the development will deliver approximately 1,500 dwellings will be located at Rowborough, 500 new homes at an extension to South Marston and approximately 6,000 dwellings south of the A420.

5.3.50 In relation to Transport the SPD identifies the interventions identified with policy NC3 of the Local Plan, viz:

- Improved White Hart junction
- Improvements to the existing network including A420
- A green bridge
- Park and Ride site
- Rapid transit link and additional public transport services
- Measures to minimise rat running through existing communities
- New road link to Commonhead

5.3.51 The SPD sets out a preferred long term solution for the Improvement of the White Hart roundabout which is described as the diamond interchange. Further detailed design and the securing of funding is required. The SPD recognises nevertheless that the Eastern Villages may proceed by means of the “phased implementation of improvements”.

5.3.52 With regard to the A420, the improvements required are, unsurprisingly, consistent with those set out in the Local Plan and South Marston SPD (see below) and include:

- a new eastern access to Rowborough (north);
- an eastern access to the Park and Ride site in the south
- improvements to the junction at Old Vicarage Lane;
- a new access to serve the new employment area (to the south);
- improvements to the Police Station junction and improvements to Gable Cross junction.

South Marston Draft Supplementary Planning Document(June 2003) and Background Transport Assessments

5.3.53 The purpose of the SPD was to provide a planning framework setting out the expectations and requirements for the sustainable development of the village of South Marston. While it is no longer being progressed beyond the draft stage, it is still referred to in the adopted Local Plan as a key evidence base document. In preparation of the SPD, SBC commissioned JMP to investigate the existing challenges and to review the potential impact of the Eastern Villages.

5.3.54 The JMP initially identified a range of potential issues with in the transport system together with possible interventions, as detailed below.

Description	Constraint	Intervention
Pedestrian and vehicular conflict around Pound Corner	Insufficient carriageway and footway available	Realignment of the carriageway at Pound Corner to provide footway on the northern side of Old Vicarage Lane
Gaps in footway provision	Thornhill Road close to the industrial estate. Vicinity of South Marston Hotel	Continuous footway from Pound Corner to Keypoint roundabout and from Pound Corner to South Marston Hotel
Accessibility improvements	Lack of Dropped Kerbs / Tactile paving	To provide dropped kerbs / tactile paving in locations where these

		are not provided
High Traffic Speeds/ Through Traffic	Little evidence to support view in terms of speeds/accidents or traffic volumes.	20mph zone for village centre & 30 mph between village centre and A420 for consistency. Some additional traffic calming measures

5.3.55 The JMP study then investigated the potential impact of the delivery of the Eastern Village concentrating on the impact on the A420 and within South Marston. This report notes that increase in traffic arising from development are from a very low base. The report concluded that "with the exception of the junctions along the A420, junctions within South Marston are expected to operate with minimal queuing. This assumes that a package of measures is in place for the A420 and A419 which deters vehicles from rat running through the village" (JMP Technical Note: "South Marston Supplementary Planning Document Part Three Transport Assessment", June 2012". Rat running is therefore avoided, on JMPs analysis, providing journey times along the A419 and A420 continue to make these routes attractive to use. The junctions with the A420 are identified as needing interventions to ensure that this is the case.

5.3.56 A key conclusion of the JMP technical note, reinforcing the conclusion regarding transport impacts in South Marston itself was that a roundabout junction on Old Vicarage Lane – providing access from Rowborough would be "unlikely to result in a significant increase in traffic volumes through South Marston village"(para 5.6) since most traffic is likely to travel south.

5.3.57 The study also identified options for improvements at Pound corner involving one or two way traffic and seeking to widen the existing footways noting that there were pros and cons of each alternative.

5.3.58 The purpose of the South Marston Supplementary Planning Document (SPD) was to provide a detailed planning framework for South Marston, setting out the expectations and requirements of development within the village. It addresses the existing and the expanded village. The SPD described a vision that: "South Marston should be a high quality, integrated village with an enhanced sense of community,

activity and safety within a distinctive form that builds on the existing character and features in a sustainable manner". In order to deliver the vision a number of issues will need to be addressed including, in relation to transport: "improvement of existing, and provision of new highway infrastructure, including resolving issues of through-traffic, Pound Corner, the absence of road-side paths, access onto the A420 and off-site provisions to ensure free movement along the A420"(para 3.3).

5.3.59 The Development Principles in respect of Access and Movement are set out in para 6.28 to 6.46. The SPD requires:

- a Transport Assessment (TA), to be undertaken by the developer(s), to determine the number of additional trips generated and how these will be accommodated on the existing highway network; and
- expects a sustainable Travel Plan to accompany the TA to promote alternatives to single occupancy car use.

The SPD promoted:

- A 20mph speed restriction throughout the village (30mph outer limit) with psychological traffic calming and new gateway features;

- The retention of Rowborough Lane and Nightingale Lane as public rights of way/bridleways, with no new development being served by either;

- Junction improvements on A420 (including at Gablecross and at the Police Station access, with the introduction of a new eastern access near Acorn Bridge and capacity improvements along the A420;

- Improvements to Pound Corner including for instance the provision of a raised footway on the northern side of Pound Corner;

- An all movement junction to link Rowbrough to Old Vicarage Lane is proposed consistent with the advice of JMP;

- The re-alignment and signalisation of the junction of Old Vicarage Lane and the A420 in order to provide a safe egress onto the A420, particularly for those vehicles travelling west towards White Hart Junction;

- The design of highways as a series of streets with important public realm functions, rather than as roads whose primary purpose is the movement of traffic;

- Enhanced footpath and cycle connections.

5.3.60 The highway principles and technical work determine that there is no need of a n eastern bypass for South Marston and that connections between Old Vicarage Lane and Thornhill Road should not be considered a “southern bypass” for the village

SBC Local Transport Plan 3 (LTP3)

5.3.61 The Local Transport Plan 3 was adopted in March 2011 and covers the period from 2011- 2026. It sets out an overarching strategy for the delivery of the Council’s transport programme. The mission statement for the LTP3 is:

> “a safe, effective and fit for purpose transport network that supports Swindon’s ambitions for town centre regeneration and economic growth whilst protecting and enhancing quality of life and the environment for the benefit of local residents, visitors and businesses”

5.3.62 While LTP3 covers maintenance and small schemes which are not identified within the IDP, larger, more strategic transport schemes are also included.

5.4 BASELINE CONDITIONS

Highway Network

5.4.1 The majority of the road network adjacent to the Site is managed by SBC, with the A419 and the M4 Motorway being managed by the HA. The roads managed by the HA are typically identified as being a part of the Strategic Road Network (SRN).

5.4.2 The A420 is located to the south of the Site and forms an important corridor within the local road network. The A420 commences from Swindon and heads generally in a north easterly direction extending into Oxford. The A420 is a predominantly single carriageway road with the national speed limit applying. There are sections of dual carriageway and the section approaching Swindon is subject to a 50mph speed limit.

5.4.3 There are several junctions along the length of the A420 which are predominantly at-grade. The junctions along the A420 that are in the environs of the Site are:

- A420 / Old Vicarage Lane
- A420 / Police HQ
- A420 / Gablecross
- A420 / A419

5.4.4 The A420 forms a priority T junction with Old Vicarage Lane and provides a ghost island right turn bay for vehicles travelling along the A420. From the junction with the A420, Old Vicarage Lane continues in a north westerly direction towards the village of South Marston. Old Vicarage Lane is rural in character and does not provide any dedicated facilities for pedestrian or cyclists. Around 100m north of the A420, Old Vicarage Lane passes under the Swindon to Paddington rail line.

5.4.5 Old Vicarage Lane forms simple priority junctions with Nightingale Lane and Rowborough Lane. These are minor roads which provide access to a small number of existing residential properties.

5.4.6 Around 1.4km to the west of the junction with Old Vicarage Lane, the A420 forms a signalised junction with the access to the Wiltshire Police headquarters. There is no facility to undertake a right turn from the A420 into the police headquarters so those

vehicles needing to access this facility need to continue through this junction towards the roundabout junction at Gablecross to undertake a U-turn. Heading in a westerly direction from this junction, there is an off road segregated walking and cycling route.

5.4.7 The Gablecross junction is around 200m to the west of the access to the Police HQ and is a five arm roundabout. The arms of the roundabout serve a supermarket to the south, Thornhill Road and the St Margaret's Retail Park to the north, with the remaining arm being exit only from the retail park. The off road pedestrian/cycle route continues westerly through the Gablecross junction. There is an unsignalised crossing of the Thornhill Road arm with a signalised crossing to the west of the junction to provide a safe route into the supermarket.

5.4.8 Thornhill Road heads in a generally northerly direction to access South Marston village. Thornhill Road is rural in nature and does not provide for dedicated facilities for pedestrian or cyclists within the highway over much of its length. There is however a pedestrian and cycle route that is segregated from Thornhill Road between the Gablecross roundabout and South Marston village. There is traffic calming (single file traffic) at the entry and exit to the village which attempts to control vehicle speeds into South Marston.

5.4.9 Thornhill Road continues into the village of South Marston where there are sections of a discontinuous footway. In the centre of South Marston, Thornhill Road forms a priority T junction with Old Vicarage Lane.

5.4.10 Around 300m to the west of the Gablecross junction lies a grade separated junction between the A419 and the A420, known as the White Hart interchange. The interchange has two parts that are connected by part of Merlin Way. The northern part of the interchange is a grade separated roundabout with the A419 main carriageway passing over the roundabout. The roundabout connects the A420, the A419 southbound slip roads on the east side of the roundabout together with Oxford Road and Ermine Street to the west and Merlin Way to the south. The northbound A419 on and off slips have a roundabout junction with Merlin Way that is located 400m to the south of the White Hart roundabout junction and form the southern part of the interchange. Any vehicles going to the north on the A419 from the A420 have to travel southwards along Merlin Way before travelling north along the A419. Oxford

Road continues from the White Hart interchange westerly towards Swindon town centre.

5.4.11 Merlin Way continues south from the White Hart interchange and becomes Wanborough Road which crosses over the A419 via a single carriageway road bridge 1.5 Km south of White Hart Roundabout. Wanborough Road links with Wanborough and Upper Wanborough, and has one lane per direction.

5.4.12 Continuing from the White Hart interchange, Oxford Road (A4312) provides a westerly link into Swindon town centre. Oxford Road is a single carriageway road with a footway to the north and a footway/cycle way to the south. Oxford Road is a residential road allowing direct access to individual properties. Oxford Road forms a priority junction with Wanborough Road that includes a signalised pedestrian crossing, a signalised junction with Nythe Road and a roundabout junction with Dorcan Way, with this junction being identified as the Greenbridge roundabout.

5.4.13 From the White Hart interchange the A419 heads in a north-westerly direction to Cirencester and in a southerly direction to link with the M4 at junction 15. The A419 takes the form of a lit dual-carriageway which is subject to a national speed limit of 70mph. There are no pedestrian or cyclist facilities along the A419.

5.4.14 The M4 motorway caters for long distance east – west movements and provides the major link between South West England / Wales to the west and Reading / London to the east. The grade separated junction with the M4 is signalised. The roundabout is lit and there are no pedestrian or cyclist facilities at the junction.

Pedestrians and Cyclists

5.4.15 The Site does not contain any significant traffic generators such that there has been no requirement for significant footway provision on site or to access the Site.

5.4.16 The primary roads that do provide footways have been identified earlier in this section.

5.4.17 Swindon has a cycling network consisting of a mixture of both on and off road routes that also caters for walking trips. As indicated earlier an off road cycling route

commences from the police headquarters and continues westerly towards the town centre. A cycle way extends northwards from the junction with Gablecross to link with South Marston. There are additional routes to the east of the Site that utilise Nightingale Lane and Rowborough Lane.

5.4.18 The desire line from the Site is predicted to be westerly towards the Sainsbury supermarket south of Gablecross and the town centre. As indicated above there is a walking and cycling link along the Thornhill Road corridor. This provides a continuous link to the Sainsbury supermarket and to the White Hart interchange. The interchange provides a pedestrian bridge for onward travel towards the town centre.

Public Rights of Way

5.4.19 Public Rights of Way (PRoW) are classified as highways and as such are protected routes. The 1949 National Parks and Access to the Countryside Act placed a duty on every County Council in England and Wales to draw up and publish a definitive map and statement of PRoW in their area.

5.4.20 The Definitive Map is the legal record of the location and status of PRoW. The statement is a description of the PRoW shown on the definitive map.

5.4.21 There are four classifications of PRoW:

- footpaths - by foot only
- bridleways - by foot, horse or bike
- restricted byways - by any form of non-motorised transport
- byways open to all traffic - by any form of transport, including cars

5.4.22 There is a network of bridleways and footpaths crossing the Site that connect with bridleways that penetrate other roads in the vicinity of the Site. In particular there are existing strong links between South Marston and the A420 for pedestrians and cyclists to Gablecross and for pedestrians across the railway at underbridges further to the east near the Police HQ and to the west and east of Old Vicarage Lane.

Road Based Public Transport

5.4.23 A range of bus services operate within Swindon, with services 65 and 66 operating closest to the Site.

5.4.24 The bus service 65 is a route managed by Stagecoach and operates between Swindon and Faringdon, Monday to Saturday. The first bus leaves Swindon Bus Station at 10:50 calling at South Marston at 11:03. For trips towards Swindon, the first bus calls at South Marston at 10:26 with a journey time of 15 minutes. This service operates typically four times daily.

5.4.25 The Route 66 is managed by Stagecoach and operates along the A420 between Swindon and Faringdon, Monday to Sunday. The first bus leaves Swindon Bus Station at 05:40 and typically operates on a half hourly frequency. The journey time to Oxford is 1 hour and 15 minutes.

Public Transport - Rail

5.4.26 The closest train station is located in central Swindon, around 6km to the west of the Site. The train station, managed by First Great Western, provides a range of facilities including:

- 578 space car park open 24 hours
- Ticket office open 7 days
- Self-service ticket office
- Manned help desk
- Cash machine
- Pay phones
- Post box
- Public WiFi
- Refreshments
- Shops
- Toilets
- Waiting rooms

5.4.27 Swindon train station provides three routes per hour to Didcot with a journey time of around 17 minutes; from Didcot, onward travel to Oxford is possible. Additional

routes to London Paddington are provided three times per hour with a journey time of 63 minutes.

5.4.28 Trips westwards to Bristol are catered for, with two trips per hour each to Temple Meads (45 minute journey time) and to Bristol Parkway (27 minutes).

Accident Review

5.4.29 Data was obtained from Wiltshire Police pertaining to all personal injury road accidents (PIAs) reported as occurring during the five year period up to the end of February 2013 on the A419, A420, A4312 and A361 in the vicinity of the Site.

5.4.30 A total of 420 accidents were reported, resulting in 600 casualties. Two accidents resulted in fatal injury, in both cases to a motorcyclist. A further 51 resulted in serious injury.

Table 5.4: Total number of PIAs by year and severity, with casualties

	Number of PIAs				Casualties
	SLIGHT	SERIOUS	FATAL	TOTAL	
Year 1 to end Feb'09	86	13	0	99	133
Year 2 to end Feb'10	75	14	0	89	121
Year 3 to end Feb'11	62	6	1	69	99
Year 4 to end Feb'12	71	11	0	82	119
Year 5 to end Feb'13	71	1	1	79	125
5 year period total	365	45	2	412	597

5.4.31 The most notable overall feature of the accidents is the high proportion of rear end shunts, making up 44% of all accidents. Also, 20% of the accidents involved vehicles skidding.

5.4.32 Other notable common factors are that 131 or 31% of the accidents involved at least one young driver or rider aged 23 years or under and that 30% of the accidents involved vulnerable road users i.e. pedestrians and the riders of 2-wheeled vehicles. The numbers of such accidents are listed below.

Table 5.5: PIAs involving young drivers/riders

	Number of PIAs				
	Under 17	17-18	19-20	21-23	TOTAL
Pedal cyclist	2	4	2	2	10
Motor cyclist	6	3	5	5	19
Car/LGV driver	0	25	33	41	99
HGV 7.5t	0	0	0	1	1
5 year period total	8	32	40	49	129

Table 5.6: PIAs involving vulnerable road users

	Number of PIAs			
	PEDESTRIAN	PEDAL CYCLIST	MOTOR CYCLIST	TOTAL
Year 1 to end Feb'09	2	10	13	25
Year 2 to end Feb'10	4	10	17	31
Year 3 to end Feb'11	2	6	6	14
Year 4 to end Feb'12	9	8	12	29
Year 5 to end Feb'13	6	7	9	22
5 year period total	23	41	57	121

5.4.33 For the purpose of more detailed analysis, the accidents have been split by area as shown in the table below and significant clusters within each area have then been identified and studied in greater detail, and compared against expected accident rates from the Cost Benefit Analysis, (COBA) from the Design Manual for Road and Bridges, (DMRB) for that particular type of road layout and volume of traffic.

Table 5.7: PIAs, by severity, split into in sub- areas for more detailed analysis

	Number of PIAs			
	SLIGHT	SERIOUS	FATAL	TOTAL
Area 1: M4 Junction 15	32	4	0	36
Area 2: A419, M4 to White Hart	27	1	1	29
Area 3: White Hart Roundabout	41	2	0	43
Area 4: A419 North of White Hart	29	7	0	36
Area 5: A420 E of A419	18	2	0	20
Area 6: A4312 W of A419	65	4	0	69
Area 7: A361 E of A419	27	0	0	27
Area 8: Other - main roads	78	20	1	99
Area 9: Other - unclassified	48	11	0	59
5 year period total	365	51	2	418

Area 1/Cluster 1 – M4 Junction 15 Interchange with A419 and A346

5.4.34 This area of the road network covers the whole of the M4 Intersection with the A419 and A346, including the immediate junction approaches and is shown as “Cluster 1” on Figure 1 within **Appendix 7a**.

5.4.35 A total of 36 PIAs were reported at this location during the 5-year study period, equivalent to 7.2 PIAs per annum. Of these none resulted in fatal injury but 4 resulted in serious injury, 3 to motorcyclists. These serious accidents do, however, appear to be on a falling trend with 2 occurring during the first 12 month period and 1 in each of the two subsequent years and none in either of the most recent two years.

5.4.36 There were a total of 52 casualties, averaging 1.4 per PIA.

5.4.37 The number of accidents per annum expected at a junction of this type and traffic level based upon COBA parameters would be 7.3 PIAs per annum which compares with an observed rate of 7.2. This suggests that the junction is performing better than might be expected and that overall the results reported give no particular cause for concern.

5.4.38 The only notable feature of the accidents is that 25 or 67% of them involved rear end shunts, 13 of these occurring on the A419 southbound approach and entry to the roundabout.

Area 2 – A419 between M4 Junction 15 and White Hart Roundabout

5.4.39 This section of the road network covers the whole of the A419 between the M4 motorway to the south and White Hart Roundabout to the north.

5.4.40 A total of 29 PIAs were reported at this location during the 5-year study period, equivalent to 5.8 PIAs per annum. Of these one resulted in fatal injury and eight resulted in serious injury. The accidents do, however, appear to be on an overall falling trend with 6, 7, 10, 2 and 4 occurring in each of the 12-month periods studied.

5.4.41 The number of accidents per annum expected along a road of this type and traffic level based upon COBA parameters would be 0.141 PIA/mvkm or based on “Road Casualties GB 2012 Table 10003” would be 0.177, which compares with an observed rate of 0.067. The overall accident record, therefore, would appear to be relatively good.

5.4.42 Notable factors amongst the 29 reported accidents are that 55% of them involved rear end shunts and 52% involved at least one vehicle skidding. 10% involved HGVs.

5.4.43 One fatal accident occurred in April 2010 just north of Commonhead Roundabout when a pedal cycle travelling south with no lights was hit by a heavy goods vehicle (HGV), displacing the rider who was then run over by the HGV and three following

vehicles. The fact that the rider was wearing dark clothing and displaying no lights is mentioned as a contributory factor in the accident although lighting conditions were not given as being "dark".

Area 3 – A419 junction with A420 White Hart Roundabout

5.4.44 At this Intersection 43 PIAs were reported including two that resulted in serious injury.

5.4.45 The number of accidents per annum expected at a junction of this type and traffic level based upon COBA parameters would be 4.6 PIAs per annum which compares with an observed rate of 8.8. This suggests that the junction is performing worse than might be expected in terms of safety.

5.4.46 By far the most notable pattern amongst these accidents is that 88% of them involved rear end shunts, mainly at the three major entries to the roundabout:

- From A419 southbound off-slip – 15 PIAs, all involving rear end shunts,
- From A420 westbound – 16 PIAs, all but one involving rear end shunts,
- From A420 Merlin Way – 9 PIAs, all but one involving rear end shunts.

5.4.47 The two principal causation factors given for these accidents are, "failed to look properly" and "failed to judge other persons path or speed". There is clearly an issue at this junction concerning drivers not slowing sufficiently on the approaches and/or judging correctly when it is safe to proceed.

5.4.48 Proposals outlined elsewhere in this report to signalise the entries onto White Hart Interchange, where this problem currently exists, is likely to have safety benefits in that with a more formalised junction control drivers' decisions are made easier and there is less room for misjudgement to occur. The detailed design process will consider the accident history of this junction and will include any necessary features.

Area 4 – A419 between White Hart Roundabout and Blunsden

5.4.49 This section of the road network covers the A419 between White Hart Roundabout and Blunsden to the north.

5.4.50 A total of 36 PIAs were reported at this location during the 5-year study period, equivalent to 7.2 PIAs per annum. Of these seven resulted in serious injury. Although spread over the length of road in question, five of these accidents did occur during the hours of darkness and three involved HGVs. The accidents do, however, appear to be on an overall falling trend with 12, 9, 5, 7 and 3 occurring in each of the 12-month periods studied.

5.4.51 The number of accidents per annum expected along a road of this type and traffic level based upon COBA parameters would be 0.141 PIA/mvkm or based on "Road Casualties GB 2012 Table 10003" would be 0.177, which compares with an observed rate of 0.076. The overall accident record, therefore, would appear to be relatively good.

5.4.52 Notable factors amongst the 36 reported accidents are that 56% of them involved rear end shunts and 42% involved at least one vehicle skidding. 22% involved HGVs.

Area 5 – A420 east of White Hart Roundabout

5.4.53 This section of the road network covers 3.0km of the A420 between the A419 White Hart Interchange and the county boundary to the east.

5.4.54 A total of 20 PIAs were reported during the 5-year study period, equivalent to 4.0 PIAs per annum. Of these none resulted in fatal injury but two resulted in serious injury.

5.4.55 The number of accidents per annum expected along a road of this type and traffic level based upon COBA parameters would be 0.309 PIA/mvkm or based on "Road Casualties GB 2012 Table 10003" would be 0.177 which compares with an observed rate of 0.168. The overall accident record, therefore, would appear to relatively good.

5.4.56 Notable factors amongst the 20 accidents are that 35% of them involved rear end shunts and 35% occurred on a wet road surface. Seven of the accidents involved vulnerable road users: one pedestrian, four pedal cycles and two motorcycles.

5.4.57 Three small clusters have been identified on this section of road. They are small but in view of their proximity to the development site they have been analyses separately in order to identify any addressable issues. These are shown on Figure in Appendix B as Clusters 1, 2 and 3.

5.4.58 **Cluster 1 – A420 Gablecross Roundabout:** Four accidents were reported at Gablecross Roundabout, one of which resulted in serious injury to a pedal cyclist when a car entered the junction from Thornhill Road, failing to give way to the pedal cycle, which was already on the roundabout. Another of the accidents involved a pedal cyclist travelling west who failed to give way to a car already on the roundabout. Another accident involved a motorcyclist travelling east who hit a car whilst changing lane on the exit from the roundabout. The fourth one involved and was caused by an unmarked police car also changing lane on the eastbound exit. There is a cluster of accidents at this junction that involved vulnerable users. This junction has been identified to be signalised that will include controlled crossing points that will mitigate any increased risk of accidents.

5.4.59 Although three of the four accidents involved vulnerable road users there are no clear common factors to suggest any particular problem.

5.4.60 The number of accidents per annum expected at a junction of this type and traffic level based upon COBA parameters would be 1.4 PIAs per annum which compares with an observed rate of 0.8 which is considerable lower than this.

5.4.61 **Cluster 2 – A420 in vicinity of Esso Service Station:** Five slight injury accidents were reported in the vicinity of the Esso service station between Gablecross Roundabout and the priority junction with Vicarage Lane. One of the accidents involved a single vehicle losing control on an icy road surface. Three of the others involved a rear end shunt in slow moving traffic, two eastbound and one westbound. The fifth accident involved a vehicle overtaking a slow moving vehicle and hitting a tractor-with-trailer turning right ahead of it.

5.4.62 None of the accidents appear to have resulted directly from activity associated with the filling station but all are a by-product of traffic slowing down in the general vicinity.

5.4.63 **Cluster 3 – A420 junction with Vicarage Lane:** Three slight injury accidents were reported at the junction of Old Vicarage Lane with the A420. One of the accidents involved a rear end shunt on the eastbound approach to the junction. The other two both involved single vehicles, one losing control turning left into Old Vicarage Lane on an icy road surface and the other whilst swerving to avoid an animal in the road. These latter two accidents resulted from unusual conditions and there does not appear to be a problem specifically related to this junction.

5.4.64 The number of accidents per annum expected at a junction of this type and traffic level based upon COBA parameters would be 0.7 PIAs per annum, which compares with an observed rate of 0.6, indicating that the junction is performing within acceptable parameters.

Area 6 – A4312 and A4259 west of A420

5.4.65 This 3.5km section of the road network covers the A4312 between the A419 in the east and Drakes Way in the west, and the A4259 just south of the A4312.

5.4.66 A total of 69 PIAs were reported at this location during the 5-year study period, equivalent to 13.8 PIAs per annum. Of these none resulted in fatal injury but four resulted in serious injury. These serious accidents do, however, appear to be on a falling trend with one, three, zero, zero and zero in each of the 12-month periods.

5.4.67 The number of accidents per annum expected along a road of this type and traffic level based upon COBA parameters would be 0.717 PIA/mvkm or based on “Road Casualties GB 2012 Table 10003” would be 0.537 which compares with an observed rate of 0.530. The overall accident record, therefore, would appear to relatively good.

5.4.68 Notable factors amongst the 69 reported accidents are that 46% of them involved rear end shunts and 35% involved drivers/riders aged 23 years of age or under. 29% of the accidents involved vulnerable road users: two pedestrians, six pedal cycles and 12 motor cycles. Further analysis highlights that 11 of the accidents involved both at least one young driver/rider and at least one vulnerable road user. The most notable common factor was 7 accidents involving motor cyclists aged 23 or under.

5.4.69 There are, however, no notable clusters of accidents along the route or other common characteristics indicating specific problem areas or addressable contributory factors.

Area 7 – A361 east of the A419

5.4.70 This section of the road network covers the A361 between the A419 and 3.0km to the east.

5.4.71 A total of 27 PIAs were reported at this location during the 5-year study period, equivalent to 5.4 PIAs per annum. Of these none resulted in either fatal or serious injury.

5.4.72 The number of accidents per annum expected along a road of this type and traffic level based upon COBA parameters would be 0.309 PIA/mvkm or based on "Road Casualties GB 2012 Table 10003" would be 0.177 which compares with an observed rate of 0.250. The overall accident record, therefore, would appear to fall somewhere between the two and does not give any particular cause for concern.

5.4.73 The only notable factor amongst the 27 reported accidents are that 37% of them involved vulnerable road users: zero pedestrians, four pedal cycles and six motor cycles. There are no notable clusters of accidents along the route.

Area 8 – Other main roads

5.4.74 This area covers all main roads of the study area that are not included in the sections above.

5.4.75 A total of 99 PIAs were reported on these roads during the 5-year study period, equivalent to 19.8 PIAs per annum. One accident resulted in fatal injury and 20 in serious injury.

5.4.76 Notable factors amongst the 99 reported accidents are that 44% of them involved vulnerable road users: nine pedestrians, 14 pedal cycles and 17 motorcycles.

5.4.77 One fatal accident occurred in July 2012 on B4141 Kingsdown Road when a van travelling north west from the college entrance, turned left and then across the main road into the path of a motorcyclist travelling south west, displacing the rider who sustained fatal injuries.

5.4.78 There are no notable clusters of accidents on these roads and they are not likely to be subjected to any significant increase in traffic as a result of the development.

Area 9 – Other unclassified roads

5.4.79 This area covers all other roads of the study area that are not included in the sections above.

5.4.80 A total of 59 PIAs were reported on unclassified roads during the 5-year study period, equivalent to 11.8 PIAs per annum. No fatal accidents were reported but eleven resulted in serious injury, eight of these occurring between 8pm and 3am.

5.4.81 Other notable factors amongst the 59 reported accidents are that 51% of them involved vulnerable road users: nine pedestrians, 12 pedal cycles and nine motorcycles. This high percentage of accidents involving vulnerable road users is indicative of the mixed use of these roads by different types of road user, for a multitude of local and longer distance activities.

5.4.82 There are no notable clusters of accidents on these roads and they are not likely to be subjected to any significant increase in traffic as a result of the development.

Injury Accidents Involving Vulnerable Road Users

5.4.83 The initial analysis of road injury accident data that was carried out for the study highlighted the fact that there were several accidents that involved vulnerable road users. This section reviews in more detail those accidents, to determine whether any significant common factors exist. This may help identify any particular remedial intervention that might ameliorate the situation.

Pedestrian Accidents

5.4.84 23 accidents resulting in injury to pedestrians were reported during the five-year study period (6% of the total). None of the accidents resulted in fatal injury but ten of the 23 (43%) resulted in serious injury. The locations of the accidents are shown in the Appendix.

5.4.85 The number of pedestrian accidents per year shows a generally upward trend, with 2, 4, 2, 9 and 6 in each of the five years respectively. The most notable feature of the accidents is that 7 (30%) involved children, 4 aged under 11 years and 3 aged 12-15 years. The only other notable feature is that 8 (35%) of the accidents occurred during the evening, between 7pm and 11pm.

5.4.86 One notable observation is that four of the accidents occurred after a driver had exited their vehicle (and therefore become a pedestrian) following another incident.

5.4.87 The locations of the accidents are shown in the Appendix.

5.4.88 The only significant cluster is five accidents in the vicinity of Kingsdown School in Upper Stratton. Four of these involved children, aged 10, 12, 13 and 15 crossing the road and failing to give way to oncoming traffic, with three resulting in serious injury. The locations, again, are fairly disperse and no notable site-specific problems are indicated by the data.

Pedal Cycle Accidents

5.4.89 41 accidents resulting in injury to pedal cyclists were reported during the five-year study period, 10% of the total. One of the accidents resulted in fatal injury when a pedal cyclist without lights travelling south in the dark on the A419 just north of Commonhead was hit by a lorry. Another six accidents resulted in serious injury. The locations of all of the accidents are shown in the Appendix.

5.4.90 The accidents appear to be slightly downward trending but there no obvious common or significant factors. There do, however, appear to be a number of small clusters, each of which is discussed in more detail below.

5.4.91 **Cluster 1 : Four accidents on B4006 in vicinity of Kingsdown School** - This cluster is not particularly significant other than it is at the same location as the clusters of pedestrian accidents that has already been identified. Four accidents were reported, one of which resulted in serious injury. Only one of the cyclists was of school age i.e. 16. There are no clear common factors amongst the reported characteristics of the accidents. The analysis indicates that the majority of the accidents did not include school age children and there was less than one accident per year. The delivery of the development is unlikely to alter the characteristics of the traffic flow such that an increase in accidents is likely.

5.4.92 **Cluster 2 : Five accidents in vicinity of A4312 Drakes Way junction with Queens Drive** - A further two accidents which initially appear to be part of this cluster are in fact on a parallel side road, and have been excluded from this analysis. Four of the remaining accidents occurred in fairly close proximity but with no other common characteristics. It is noted, however, that 2 of the cyclists were cycling on the pavement. No children (aged 16 and under) were involved in the accidents reported at this location.

5.4.93 **Cluster 3 : Five accidents on A4312 Oxford Road south west of White Hart Roundabout** - The five accidents occurred at different locations under different circumstances. It is noted, however, that 3 of the cyclists were cycling on the pavement. No children (aged 16 and under) were involved.

5.4.94 **Cluster 4 : Four accidents on A420 east of White Hart Roundabout** - The grid references given for two of these accidents place them at White Hart Roundabout but in fact the descriptions all mention Gablecross. No young cyclists were involved and in two of the accidents the cyclist was at fault. In the other two, including the serious injury accident, a vehicle entering the roundabout failed to give way to a cyclist already on the circulating carriageway. Three of the accidents occurred on a Monday although it is difficult to attach any significance to this. Three occurred between 1630 and 1800 in the evening but only two of these on weekdays, one on a Saturday. Again, there are no clear patterns indicating specific problems.

5.4.95 **Cluster 5 : Four accidents at the Merlin Way roundabout with Covingham Drive** - All four accidents here occurred as a result of a vehicle pulling out of Covingham Drive and failing to give way to a cyclist entering the roundabout from Merlin Way

south. Three of these occurred during a weekday morning peak hour when drivers may have been looking into the sun (one accident mentions dazzling as a possible contributory factor). Otherwise, visibility is good and generally free of obstructions.

Vulnerable Road User Summary

5.4.96 A more detailed analysis of road injury accidents involving pedestrians and pedal cyclists has identified a number of clusters of accidents but has failed to determine any particular common factors that indicate any specific problems that might be susceptible to remedial intervention.

5.4.97 There have, however, been a total of 9 accidents in 5 years involving vulnerable road users in the vicinity of Kingsdown School in Upper Stratton. Five children (aged between 10 and 16 years), averaging one per year, were injured on the roads surrounding the school during the study period.

Accident Summary

5.4.98 418 personal injury accidents were reported to have occurred within the study area during the most recent 5-year period for which information is available at the time of writing. This included two fatal accidents, both involving riders of two-wheeled vehicles. Overall there does appear to be a quite high proportion of rear shunt type accidents, and in several cases, accidents involving vulnerable road users, although this is not necessarily atypical of urban roads serving a mixture of purposes in a living environment.

5.4.99 The area has been split into several sections for further analysis and the numbers of accidents reported has been compared with statistics indicating expected rates for each type of road or junction given observed traffic levels. Overall the accident record appears to be quite good with observed figures generally below the equivalent expected rate.

5.4.100 The one notable exception to this is at White Hart Roundabout where a disproportionate number of rear shunt type accidents have occurred on three of the approaches to the junction. Proposals outlined elsewhere in this report to improve the capacity of this junction are likely to largely address this problem in the future by

introducing signal control on two of the entries. These proposals will formalise the operation of junction through the introduction of traffic signals and therefore remove the decisions associated with drivers having to give way, simplifying their thought processes and thereby reducing errors of judgement.

5.4.101 To summarise, there are no notable safety issues that will remain unaddressed by the current development proposals. Although the development will add traffic to the network there is no reason to suppose that his will significantly compromise the relatively safe performance of the existing road system.

Base Line Traffic Levels

5.4.102 To establish the existing traffic conditions within Swindon, traffic flow data was extracted from the Paramics and Saturn traffic models. The impact of the Proposed Development was assessed against the period of peak traffic, typically between 08:00 – 09:00 and 17:00 – 18:00.

Table 5.8: Traffic Levels

Highway Link	Base year		2026 Future year	
	AM Peak	PM Peak	AM Peak	PM Peak
M4 Motorway	6444	6745	7374	7813
A419, J15 to Commonhead	5531	5399	4159	3837
A4259 at Commonhead junction	2825	2966	3339	3200
Purley Rd at Commonhead junction	403	600	595	480
Pack Hill at Commonhead junction	251	309	210	117
A419, Commonhead to White Hart	5084	4665	5715	5393
A420 at White Hart	2753	3218	2930	3051
Ermin Street at White Hart	533	674	703	835
Oxford Road at White Hart	1410	1309	1440	1277
Merlin Way at White Hart	1400	1458	1850	2114
A419 north of White Hart	5350	5501	6277	5845
Thornhill Road north of Gablecross	413	405	530	645
A420 east of Gablecross	2064	2055	2136	2072
Supermarket/access S of Gablecross	479	745	2737	2868
A420 west of Old Vicarage Lane	2163	2165	541	806
A420 east of Old Vicarage Lane	1969	1916	2081	2026
Old Vicarage Lane north A420	157	132	106	66
Old Vicarage Lane S of Highworth Rd	124	94	101	66
Thornhill Road west of Highworth Rd	280	209	500	391
Highworth Rd N of Old Vicarage Lane	282	209	0	0
A420 west of New Eastern Access	1969	1916	1986	2011
A420 east of New Eastern Access	1969	1916	1984	2007
New Eastern Access Road	0	0	0	0
Link between Honda RB and Hyde Rd	4595	4536	5633	5041
Link between Hyde Road and Bypass	4585	4484	5546	4927

5.5 POTENTIAL EFFECTS

Construction stage

5.5.1 It is anticipated that construction activities will be undertaken over a period of up to some 12 years and due to the complexity and length of the construction programme it is not possible to accurately predict volumes of traffic that will be generated over the course of a normal working day. However, a qualitative assessment can be carried out as described below.

5.5.2 If it is assumed that some 300 dwellings were constructed in a year then it is likely that this could result in circa 75 dwellings being in the process of being constructed at any one time. This could result in around 75 tradesmen being on site at any one time which would lead to 150 two way trips per day. It is assumed that there would be in the region of up to 10 HGV movements per day from vehicles accessing the Site, which would lead to 20 two way trips per day.

5.5.3 A routing strategy is suggested which would mean that only roads best suited for construction traffic would be affected. It has therefore been assumed that all construction traffic will route via the A419 and the A420 with all other roads, including those through the centre of South Marston, prohibited to construction traffic.

5.5.4 It is considered that the effect of construction traffic on the surrounding highway network will be of no greater than minor adverse significance as the HGV movements will be scheduled to avoid the peak times of travel demand and the traffic generated by the tradesman will not be discernible from general traffic. Furthermore, the effect of the construction traffic will be minimised as construction trips will not be routed along local roads and routes that are not designed to cater for such traffic. If construction exceeded these assumptions then the impacts would still be indiscernible from the current levels of traffic.

Post-completion stage

5.5.5 Full details of the predicted traffic generation are contained within the TA.

Proposed Access Strategy

5.5.6 The Phase I Proposed Development comprises of two parts, being the growth of South Marston and to the east, the development of a new community at Rowborough. Each area will have its own distinct identity and a strong landscape and green infrastructure framework will define and shape the development of South Marston and Rowborough as individual settlements. Particular emphasis has been placed on protecting South Marston from potential undesirable effects of the growth.

5.5.7 The application provides for some 500 dwellings at South Marston and a development that will build on the existing village character and improve links to the surrounding countryside. The layout comprises of a series of residential land parcels developed on a connected network of routes, which integrate with the existing village.

5.5.8 Provision is made for education, recreation and community facilities for the expanded village. A large area of open land between the expended village and railway line is to maintain and enhance identity of the village. Measures are promoted to limit traffic within the existing and proposed extended village in line with previously agreed principles.

5.5.9 Some 1,800 dwellings are proposed at Rowborough to form one of the larger new villages within the NEV allocation. A new primary school and mixed use local centre(s) are also proposed. The stream valley area provides a strategic landscape corridor, which will remain undeveloped, punctuated by retained hedgerows and trees. The landscape corridor will provide attractive informal open and/or amenity space and reinforce separation between South Marston and Rowborough.

5.5.10 The Development Framework Plan, sets out built development components; areas of formal and informal open space; and the alignment of the primary strategic transport routes running through the application site. Key layout principles are embedded within the Parameter Plan and illustrated through the illustrative master plan:

- The development will be comprised of walkable residential neighbourhoods around distinct character areas. Each neighbourhood benefits from access to key areas of formal and informal open space;

- An accessible and central 'Neighbourhood Hub', created through co-location of community facilities, will form the focus of activity for Rowborough and be accessibly located on the main through street. A mixed use centre is proposed to the south eastern area of Rowborough, again adjoining the main through street. The South Marston SPD sets out proposals to enhance local facilities in South Marston;

- Opportunities for strategic footpath / cycle linkages into the wider network will be exploited, connecting adjoining locations within the wider Swindon Eastern Villages to the application site;

- Legibility of the development will be ensured by developing a positive relationship between buildings, streets and spaces, with buildings fronting onto and providing opportunities for overlooking and surveillance. In addition, ensuring there is a strong definition of public (streets, opens space) and private spaces (back gardens, private driveways) making the layout legible and safe;

- A variety of landscape and development responses to the existing character of the site, including provision of a robust landscape framework including a network of formal and informal open spaces, and the creation of sports pitches located in convenient and accessible locations. At South Marston pitches are located to provide recreation opportunities for the wider community;

- The development will establish a rich and diverse green infrastructure strategy incorporating existing trees and hedgerows and connecting parks, play areas, formal sports pitches, and wildlife habitats.

5.5.11 The planning application considers a total of 2,380 new dwellings. The detailed traffic modelling, summarised in this document, assumes 2,550 dwellings, to represent a worst case (highest impact) scenario.

5.5.12 Future phases of the growth have been tested and are identified as Phase IA and Phase II in accordance with profiles of growth agreed during pre-application consultations and as set out in the draft Eastern Villages SPD.

5.5.13 Design of the transport and highway proposals is considerate to the specific concerns of the local South Marston community. The access strategy has been considered to specifically limit the increase in traffic through South Marston. The selection of a roundabout junction on Thornhill Road provides a direct link into the development to reduce any need to travel through South Marston.

5.5.14 Proposals for Old Vicarage Lane include the alteration of the priority such that traffic generated by the development is encouraged through the development. Access from Rowborough is proposed onto Old Vicarage Lane at two points with the specific aim (of the southern access in particular) to provide local and immediate access to the A420. By this means traffic generated from within Rowborough will employ the access(es) to the A420 to the south rather than find its way to Thornhill Road and Gablecross roundabout via Thornhill Road.

5.5.15 As development progresses a connection through South Marston and connections at Rowborough will be provided the routes of which have been identified by the masterplan to connect to the external public highway. As such the transport strategy route provides an equitable and effective route choice for this development. The proposed link between Old Vicarage Lane and Thornhill Road is not intended to perform a southern bypass role for the South Marston area. Indeed the link has been designed with Manual for Streets philosophy in mind and will be experienced as a street within the landscape setting or context of South Marston rather than as a highly engineered road.

5.5.16 As previously identified, the Proposed Development could deliver up to 2,380 dwellings. This quantum of development will be constructed over a number of years, potentially up to around 12. Therefore the development is likely to be built in several phases with points of access delivered when necessary.

Phase 1A

5.5.17 At the time of writing, it is envisaged that development will commence from the western boundary of the Site with access taken from Thornhill Road via a compact roundabout. It is envisaged that this could serve 200 – 300 dwellings.

Phase 1B

5.5.18 A secondary access point will be delivered to the western side of Old Vicarage Lane via a three arm roundabout. In addition to this, the horizontal alignment of Old Vicarage Lane will be altered to provide a further access into the Site. This will result in Old Vicarage Lane giving way to the main line flow which will be towards the development. This will have the benefit of reducing any desire line through South Marston.

5.5.19 As development progresses, the connection of the access points onto Old Vicarage Lane and Thornhill Road will allow completion of the 500 dwellings identified to the south of South Marston by providing a second access point for each new home in the expanded South Marston.

Phase 1C

5.5.20 Access to the Rowborough phases of development will be taken from the eastern side of Old Vicarage Lane via a roundabout identified as part of Phase 1B, and as described in the work completed by JMP to support the South Marston SPD, together with a new priority junction further to the south. The horizontal alignment of Old Vicarage Lane will be amended to result in Old Vicarage Lane to the north giving way to the main line flow, which will be towards the development.

5.5.21 A further access to the Rowborough phases of development off Vicarage Lane is required to support the progressive development of proposals from the west of Rowborough.

5.5.22 Access to Rowborough is also to be delivered along the A420 via a signal controlled priority junction. The phasing of the connections back to the A420, including the timing of the delivery of the rail under bridge at the eastern end of the site, will be a matter for discussion with SBC.

5.5.23 The access strategy together with the internal road layout has been developed specifically to minimise the traffic routing through the proposed development and the existing South Marston village. The intention is to maintain the A420 as the primary east-west corridor and providing direct and ready connection to the A420 at Thornhill Road, Old Vicarage Lane, and from the new access towards the eastern end of the

development. The main connections through the site have been designed as public transport routes providing effective public transport connectivity between South Marston and Rowborough to enhance the attractiveness of public transport for residents of the new development but also the existing community. The link between Thornhill Road and Old Vicarage Lane has been designed with Manual for Streets philosophy in mind and will be experienced as a street within the landscape setting or context of South Marston rather than as a highly engineered road.

5.5.24 If not carefully designed, Old Vicarage Lane could result in additional traffic through South Marston. To reduce the attractiveness of this route, the access strategy reduces the importance of this link by altering the priorities with the development routes. This will encourage trips direct into the development rather than South Marston as was envisaged by the JMP background studies. By providing the second access to the Old Vicarage Lane, traffic generated from within Rowborough towards Old Vicarage Lane will employ the access(es) to the A420 to the south rather than find its way to Thornhill Road and Gablecross roundabout via Thornhill Road.

5.5.25 Consideration has also been given to the wider objectives in relation to transport in South Marston as set out in the Draft SPD. The objective of a 20 MPH speed limit throughout the village is supported in principle and in practice through the design of most streets within the development areas in accordance with such speeds. Rowborough and Nightingale Lanes are retained. The aspiration in Policy RA3 of the Local Plan to secure traffic management measures in the expanded and existing village (for instance at Pound Corner) are also capable of being supported through the application proposals.

Traffic impact

5.5.26 The table below compares the predicted traffic generations for the Proposed Development with the 2026 AM and PM peak hour base flows to identify any potential sections of road where the increases in traffic may exceed the thresholds set out in the IEMA Guidance, thus requiring a detailed assessment.

5.5.27 The OPA is for up to 2,380 dwellings; however, to ensure a robust assessment, the transport assessment work has been based on 2,550 dwellings.

5.5.28 The links that connect into Pound Corner are Old Vicarage Lane south of Highworth Road (the southern arm), Thornhill Road west of Highworth Road (western arm) and Highworth Road north of Old Vicarage Lane (north-east arm)

Table 5.9: Percentage Impact

Highway Link	2026 Future year		2026 with Phase 1		Percentage difference	
	AM Peak	PM Peak	AM Peak	PM Peak	AM Peak	PM Peak
M4 Motorway	7374	7019	7813	7746	-4.8%	-0.9%
A419, J15 to Commonhead	4159	4187	3837	3817	0.7%	-0.5%
A4259 at Commonhead junction	3339	3448	3200	3248	3.3%	1.5%
Purley Rd at Commonhead junction	595	601	480	492	1.0%	2.5%
Pack Hill at Commonhead junction	210	226	117	120	7.6%	2.6%
A419, Commonhead to White Hart	5715	5921	5393	5404	3.6%	0.2%
A420 at White Hart	2930	3680	3051	3974	25.6%	30.3%
Ermin Street at White Hart	703	747	835	860	6.3%	3.0%
Oxford Road at White Hart	1440	1566	1277	1580	8.7%	23.7%
Merlin Way at White Hart	1850	2090	2114	2329	13.0%	10.2%
A419 north of White Hart	6277	6293	5845	5785	0.3%	-1.0%
Thornhill Road north of Gablecross	530	787	645	937	48.5%	45.3%
A420 east of Gablecross	2136	2723	2072	2662	27.5%	28.5%
A420 west of Gablecross	2737	3508	2868	3749	28.2%	30.7%
Supermarket/access S of Gablecross	541	584	806	957	7.9%	18.7%
A420 east of Old Vicarage Lane	2081	2251	2026	2386	8.2%	17.8%
Old Vicarage Lane north of A420	106	502	66	513	373.6%	677.3%
Old Vicarage Lane S of Highworth Rd	101	96	66	144	-5.0%	118.2%
Thornhill Road west of Highworth Rd	500	359	391	282	-28.2%	-27.9%
A420 west of New Eastern Access	1986	2115	2011	2229	6.5%	10.8%
A420 east of New Eastern Access	1984	1982	2007	2035	-0.1%	1.4%
New Eastern Access Road	0	155	0	236	n/a	n/a
Link between Honda	5633	5593	5041	5018	-0.7%	-0.5%

RB and Hyde Rd						
Link between Hyde Road and Bypass	5546	5504	4927	4935	-0.8%	0.2%

5.5.29 The information contained above indicates the locations where there is predicted to be significant increases in traffic solely as a result of the Proposed Development, i.e. that will exceed the IEMA threshold of a 10% increase along sensitive links. The following links meet that criteria:

- A420 at White Hart
- Oxford Road at White Hart
- Merlin Way at White Hart
- Thornhill Road at Gablecross
- A420 east of Gablecross
- A420 west of Gablecross
- Supermarket access at Gablecross
- A420 east of Old Vicarage Lane
- Old Vicarage Lane north of A420
- Old Vicarage Lane south of Highworth Road
- Thornhill Road west of Highworth Rd
- A420 west of New Eastern Access

5.5.30 Consequently, a detailed assessment of the potential effects of the Proposed Development along the above links has been undertaken and is reported below.

5.5.31 It is noted that two of the locations identified report a reduction in vehicles, being at Thornhill Road and Highworth Road at Pound Corner. The decrease in flows at these locations occurs due to the context of the access proposals, viz the realignment of Old Vicarage Lane which encourages trips into the development. The figure above predicts that the flows approaching Pound Corner could reduce by up to 141 vehicles.

5.5.32 In summary therefore whilst the flow of traffic on those southern sections of Thornhill Road and Old Vicarage Lane down to the A420 inevitably increase, the traffic flows between the access points and the existing village generally reduce in the morning and am peaks, following the Phase 1 development. This is likely to reflect the

arrangement of the junctions to discourage traffic movements to the north but also the choice of some existing users of the links to remain on other sections of the network rather than to pass through South Marston as a result of the additional junctions that would need to be navigated. For these and other reasons there is no material increase in traffic anticipated through Pound Corner.

Severance

5.5.33 The IEMA Guidance highlights that receptors are likely to experience significant effects in terms of severance when traffic flows change by 30% or more. The locations where such an increase is predicted are indicated below:

- A420 at White Hart (30.3%)
- Thornhill Road north of Gablecross (48.5%)
- A420 west of Gablecross (30.7%)
- Old Vicarage Lane north of A420 (677.3%)
- Old Vicarage Lane south of Highworth Road (118.2%)

5.5.34 The percentage increase reported along Old Vicarage Lane to the south of Highworth Road reflects the relatively low traffic levels along Old Vicarage Lane. Even with the inclusion of the development traffic, the traffic flows at this location are not considered significant. The development will introduce an additional 78 vehicles during the peak hour, equivalent to an increase of only one additional vehicle every 45 seconds.

5.5.35 In relation to the other roads approaching Pound Corner it is predicted that the flows are predicted to decrease, by up to 141 vehicles on Highworth Road, equivalent to a decrease of one additional vehicle every 25 seconds.

5.5.36 Along Old Vicarage Lane north of the A420, there is a significant percentage increase predicted. This increase reflects the location of the site access points and the current low traffic levels. As part of the Proposed Development, it is proposed to deliver several points of access from Old Vicarage Lane; these will include footways along the access roads and crossing points within the Old Vicarage Lane highway such that the severance is not considered significant.

5.5.37 At the A420 west of Gablecross roundabout, a minor increase in traffic levels is predicted. There is currently a signal controlled crossing point and a segregated walking / cycling route at this location. It is considered that the impact on severance at this location is negligible, due to these current facilities.

5.5.38 There is currently a segregated walking / cycling route alongside Thornhill Road. The Proposed Development will provide safe links into this route. It is considered that the impact on severance at this location is negligible, again due to these current facilities.

5.5.39 At the A420 White Hart only a minor increase in traffic levels is predicted. There is currently a pedestrian bridge at the junction that links into a segregated walking / cycling route at this location. It is considered that the impact on severance at this location is considered negligible, again due to these current facilities.

Driver Delay

5.5.40 Delays to non-development traffic can occur on the network due to the additional traffic generated by a development. The IEMA Guidance notes that these additional delays are only likely to be significant when the traffic on the network surrounding the development is already at, or close, to capacity.

5.5.41 The overall traffic levels along Old Vicarage Lane and Thornhill Road are within the theoretical highway capacity of these links and therefore it is considered that this will not create significant driver delay.

5.5.42 The traffic levels predicted at Pound Corner are within highway capacity thresholds and therefore are unlikely to increase driver delay.

5.5.43 Design of the transport and highway proposals is considerate to the specific concerns of the local South Marston community. The access strategy has been considered to specifically limit the increase in traffic through South Marston. The selection of a roundabout junction on Thornhill Road provides a direct link into the development to reduce any need to travel through South Marston. Proposals for Old Vicarage Lane include the alteration of the priority such that traffic generated by the development is discouraged from travelling towards South Marston.

5.5.44 This proposed junction arrangement was also considered to be effective in avoiding traffic passing through South Marston in the technical studies conducted by JMP as background to the draft South Marston SPD. through the development.

5.5.45 As development progresses a connection through South Marston and connections at Rowborough will be provided the routes of which have main route through the site has been identified by the masterplan to connect to the external public highway.

5.5.46 Access from Rowborough is proposed onto Old Vicarage Lane at two points with the specific aim (of the southern access in particular) to provide local and immediate access to the A420. By this means the traffic modelling work confirms that traffic generated from within Rowborough will employ the access(es) to the A420 to the south rather than find its way to Thornhill Road and Gablecross roundabout via Thornhill Road.

5.5.47 This route has been design with Manual for Streets philosophy in mind. As such the transport strategy , in that this route provides an equitable and effective route choice for this development. The proposed link between Old Vicarage Lane and Thornhill Road is therefore demonstrated not to perform a but does not form a southern bypass role for the South Marston area. Indeed the link has been designed with Manual for Streets philosophy in mind and will be experienced as a street within the landscape setting or context of South Marston rather than as a highly engineered road.

5.5.48 The potential impact of the proposals for South Marston has been fully appraised, including Pound Corner. This assessment indicates that the delivery of these proposals will not materially increase the flows through the village or impede access to the A420 for existing residents.

5.5.49 Furthermore in relation to the highway assessment, the Thornhill Road, Old Vicarage Lane and the new access have all been assessed. The results indicate that the junctions will operate satisfactorily when the development has been included. Whilst the flow of traffic on those southern sections of Thornhill Road and Old Vicarage Lane down to the A420 inevitably increase, the traffic flows between the access points and the existing village generally reduce in the morning and am peaks, following the Phase 1 development. This is likely to reflect the arrangement of the

junctions to discourage traffic movements to the north but also the choice of some existing users of the links to remain on other sections of the network rather than to pass through South Marston as a result of the additional junctions that would need to be navigated. For these and other reasons there is no material increase in traffic anticipated through Pound Corner.

5.5.50 Notwithstanding, the additional traffic generated by the development, with the improvements proposed to the Gablecross roundabout and the junction of Old Vicarage Lane, the level of queuing at Gablecross roundabout on the Thornhill Road arm is forecast to reduce compared to the no development and no improvement position and the queuing on Old Vicarage Lane will be limited and acceptable. The impact on South Marston will therefore be limited.

5.5.51 It is accepted that the flows along the A420 could give rise to potential congestion and therefore driver delay. Along the A420 corridor, a range of interventions have been identified which will improve both link and junction capacity, including capacity improvements at the Gablecross junction and partial signalisation of the White Hart interchange.

5.5.52 Taking into account these identified improvements to the highway network, the effect on driver delay as a result of the increase in traffic generated by the Proposed Development is considered to be not significant.

Pedestrian Delay

5.5.53 In accordance with the IEMA Guidance, pedestrian delay is likely to occur when traffic affects the ability of people to cross roads. There are currently low levels of pedestrian activity in the vicinity of the Site, although the level of activity is likely to increase following the delivery of the Proposed Development.

5.5.54 The desire line for pedestrians from the development is likely to be in a westerly direction. The pedestrian flow north - south along Old Vicarage Lane is likely to be insignificant. Pedestrian facilities are to be included to cater for east – west trips across Old Vicarage Lane.

5.5.55 As indicated above the traffic flows around pound corner are likely to reduce and therefore the development will not have a detrimental effect on pedestrian delay.

5.5.56 Within the A420 corridor there are already pedestrian facilities, including the bridge over the White Hart interchange. It is considered that these facilities are sufficient to cater for the expected pedestrian flow.

5.5.57 The effect of the traffic generated by the Proposed Development on pedestrian delay along these links is therefore considered to be not significant.

Pedestrian Amenity

5.5.58 In accordance with the IEMA Guidance, pedestrian amenity should only be considered significant in locations where the traffic flow is doubled. There are only two locations where such changes in traffic levels are predicted:

- Old Vicarage Lane south of Highworth Road (118.2%)
- Old Vicarage Lane north of A420 (677.3%)

5.5.59 The high percentage increase in traffic predicted is as a result of the low levels of existing traffic along Old Vicarage Lane. The predicted future levels at this location are within the theoretical highway capacity and there will be improvements to the crossing facilities. Therefore, the effect of development generated traffic on pedestrian amenity along Old Vicarage Lane is considered to be not significant.

Fear and intimidation

5.5.60 The primary factor in increasing levels of fear and intimidation for pedestrians and cyclists is high percentage changes in traffic volumes and HGVs. Due to the residential nature of the development there is not predicted to be any significant increase in the volume of HGV traffic on the network and so the magnitude of change is classified as negligible.

5.5.61 There are locations identified where traffic levels do increase beyond the 30% threshold, but as indicated above there are no locations identified where pedestrian delay would increase significantly or where amenity would reduce significantly.

5.5.62 Therefore, the effect of the Proposed Development on fear and intimidation is considered to be not significant.

Accidents and Safety

5.5.63 In accordance with the IEMA Guidance, an assessment of road safety should be considered if the character of traffic flow alters through increases in volume. The Proposed Development is not predicted to generate significant volumes of HGV traffic and the TA demonstrates that traffic is not likely to increase significantly on any links that are not designed for the predicted levels. Therefore, the Proposed Development is unlikely to produce a change in character of the traffic on the surrounding road network.

5.5.64 Therefore, the effect of the Proposed Development on accidents and safety is considered to be not significant.

5.6 MITIGATION MEASURES

Construction stage

5.6.1 It is considered that construction traffic will have no greater than a minor adverse impact. However this will be further mitigated through the production of a Construction Environmental Management Plan (CEMP). The purpose is to reduce the risk of adverse effects of construction on sensitive environmental resources and to minimise disturbance to local residents.

5.6.2 The objective is to demonstrate that appropriate checking, monitoring and audit processes will be implemented to ensure works are undertaken in an appropriate manner, together with measures to ensure that appropriate corrective actions or mitigation measures are taken.

5.6.3 The CEMP shall include:-

- Details of the approved construction traffic routes;
- The times within which traffic can enter and leave the Site;
- Specified on-site parking for vehicles associated with the construction works and the provision made for access thereto; and
- Details of the expected number of construction vehicles per day.

Post-completion stage

Highway Interventions

5.6.4 As identified above, the delivery of any substantial residential development has the potential to increase traffic levels on the surrounding road network. An assessment of the potential impacts associated with the Proposed Development has indicated that there are several locations identified within the local road network that require junction interventions. These include:

- **A420 / Old Vicarage Lane:** Shuttle signals to operate at the railway underbridge on Old Vicarage Lane linked to a signal controlled junction between Old Vicarage Lane and A420, together with a change of priorities leading into

the development on the eastern side of Old Vicarage Lane with Old Vicarage Lane becoming the minor road.

- **New Access onto A420:** New signal controlled junction at the east end of phase one to provide access from the A420.
- **Police HQ Signals:** Retain the existing layout on A420, with minor changes to the signal timings. Consider an option to open the secondary exit point onto Thornhill Road, should the Police desire
- **Gablecross Roundabout:** Conversion to a signalled crossroads (this is a revised intervention from the original proposed intervention which was localised widening on approach arms)
- **White Hart interchange:** Signalisation off several arms of the existing interchange layout.
- **Merlin Way / A419 slip road roundabout:** Signalisation off several arms of the existing interchange layout (subsequent traffic modelling of the revised development layout together with anticipated traffic growth for the Swindon Borough raised that mitigation measures were necessary at this junction location).

5.6.5 Swindon Borough Council has also provided a package of off-site improvements that are considered necessary to accommodate the wider Eastern Villages and South Marston development strategy. However BCL have reviewed this package and consider that the above proposed off-site improvements as reported at paragraph 5.6.4 will accommodate the traffic generated by the proposed development.

Walking and Cycling

5.6.6 Published good practice identifies five main requirements for pedestrian routes. Wherever possible, these requirements should be followed when planning for pedestrians within the Proposed Development:

- Convenience – follow desire lines without any undue deviation from routes;
- Connectivity – link multiple origins and destinations;
- Conviviality – be pleasant to use;
- Coherence – be made legible through paving and/or signage; and
- Conspicuousness – promote security and safety allowing pedestrians to see and be seen by others.

5.6.7 The *Guidance for Cycle Audit and Cycle Review* (The Institution of Highways and Transportation, 1998) determines five main requirements for cycle routes. Again, it is important that these requirements are recognised if the promotion of cycling to the Site as a viable and attractive alternative to car use is to be successful:

- Coherence – continuous and to a consistent standard;
- Directness – closely follow desire lines as much as possible;
- Attractiveness – in aesthetic as well as objective terms;
- Safety – designed to minimise risks for cyclists and others; and
- Comfort – well maintained smooth dry surfaces, flush kerbs and gentle gradients.

5.6.8 The Proposed Development will include numerous walking and cycling routes within the Site to provide a comprehensive route network that will comprise both on and off road routes. Highway crossing points will be designed to cater for all types of pedestrian users with the routes lit where appropriate.

5.6.9 The walking and cycling routes will connect the individual housing blocks into the primary routes through the Site that will ensure full connectively and route choice throughout the development.

5.6.10 The onsite network will connect into the external walking and cycling networks that includes the PRoW network. The predominant walking and cycling desire line towards Swindon is to be fully incorporated into the movement network for the Proposed Development. Currently there is a high quality walking and cycling link adjacent to the A420 commencing from the Police HQ junction heading in a westerly direction through the Gablecross junction to connect with the White Hart interchange. The Proposed Development will also provide a connection into the walking and cycling link adjacent to Thornhill Road, which in turn links with the A420. This will ensure that pedestrian and cycle journeys into the Sainsbury supermarket and into Swindon can be carried out safely and conveniently.

Public Transport

5.6.11 To maximise the opportunities to travel by public transport, it is proposed to improve the current routes that operate adjacent to the site. The options to deliver public transport enhancements have been discussed with SBC. Currently there are only two routes operated by Stagecoach that operate adjacent to the Site boundary.

5.6.12 Through discussions with Stagecoach, various public transport options were investigated. It was concluded that neither of the existing routes are suitable for serving the proposed development. Therefore the preferred solution is to provide a new service through the Proposed Development.

5.6.13 The Proposed Development has been assumed to commence from Thornhill Road and proceed in an easterly direction. Through the discussions with Stagecoach, a bespoke service has been identified that delivers the level of service frequency which will be attractive to future residents and serves the key destination. The bespoke service will deliver a 20-minute frequency from the outset. The service is operated as a stand-alone service, making it a great deal easier to operate reliably.

5.6.14 The service also benefits from offering a highly competitive attractive journey time to the town centre, including to the rail station, especially when end-to-end journey times are factored in. The service will operate into the evening to maximise scope for returning rail users.

5.6.15 The proposed route will operate at a 20 minute frequency Monday to Saturday and hourly on Sunday.

Travel Plan

5.6.16 To mitigate the increase in trips, a Framework Travel Plan (TP) has been produced. The TP is attached as an appendix to the ES, found in **Appendix 7b**. The TP establishes mode share targets to reduce traffic effect on the road network and encourage a modal shift towards sustainable modes of travel. These targets are based on challenging, but achievable non-car and Single Occupancy Vehicle (SOV) mode share targets. The targets are based upon current practice in the Site's

environs and have regard to the location of the Site. The targets take account of the local geography and existing transport provision.

5.6.17 Research has shown that TPs need to be managed by a travel plan coordinator, who has a clear brief with dedicated resources to manage the TP to ensure its objectives are met. It is the intention that the Travel Plan Coordinator will be in post for 5 years after 1st occupation of the Proposed Development.

5.6.18 The key to a successful TP is identifying the correct measures that will suit future residents of a development. It is unlikely that there will be sufficient attraction to a single measure; hence a combination of measures is considered the most suitable approach to pursue in this case. The Travel Plan produced to support that planning application identifies possible measures, which could include:

- Welcome Packs
- Travel induction sessions
- Car share database
- Car club

5.7 RESIDUAL EFFECTS

Construction stage

5.7.1 Taking account of the proposed CEMP, it is considered that potential minor adverse effects would be reduced to negligible level.

Post-completion stage

5.7.2 Taking account of the proposed highway interventions, walking, cycling and public transport measures, and the TP, it is considered that potential minor adverse effects would again be reduced to negligible level.

5.7.3 In the immediate environs of South Marston, the Thornhill Road, Old Vicarage Lane and the new access have all been assessed. The results indicate that the junctions will operate satisfactorily when the development has been included. Whilst the flow of traffic on those southern sections of Thornhill Road and Old Vicarage Lane down to the A420 inevitably increase, the traffic flows between the access points and the existing village generally reduce in the morning and am peaks, following the Phase 1 development. This is likely to reflect the arrangement of the junctions to discourage traffic movements to the north but also the choice of some existing users of the links to remain on other sections of the network rather than to pass through South Marston as a result of the additional junctions that would need to be navigated. For these and other reasons there is no material increase in traffic anticipated through Pound Corner. Notwithstanding, the additional traffic generated by the development, with the improvements proposed to the Gablecross roundabout and the junction of Old Vicarage Lane, the level of queuing at Gablecross roundabout on the Thornhill Road arm is forecast to reduce compared to the no development and no improvement position and the queuing on Old Vicarage Lane will be limited and acceptable.

Summary of effects

5.7.4 The effects identified are summarised in Table 5.10 below:

Table 5.10: Summary of effects

Potential effect	Significance (pre-mitigation)	Mitigation measure	Significance of residual effect
Construction stage			
Construction traffic	**Minor adverse**	CEMP	**Negligible**
Post-completion stage			
Severance	**Minor adverse**	Improved facilities within the Site and provision of appropriate linkages to the existing facilities within the Thornhill Road and Old Vicarage lane corridors	**Negligible**
Driver delay	**Minor adverse**	Junction mitigation as identified above	**Negligible**
Pedestrian delay	**Minor adverse**	Improved facilities within the Site and provision of appropriate linkages to the existing facilities within the Thornhill Road and Old Vicarage Lane corridors	**Negligible**
Pedestrian amenity	**Minor adverse**	As for Pedestrian delay above	**Negligible**
Fear and intimidation	**Minor adverse**	As for Pedestrian delay above	**Negligible**
Accidents and safety	**Negligible**	No specific mitigation required	**Negligible**

5.8 CUMULATIVE EFFECTS

5.8.1 The cumulative effects of the Proposed Development in respect of transportation have been taken into account through the inclusion of the committed schemes and allocations identified by SBC within the modelling work. However, through discussions with SBC and the HA, it was agreed that two further scenarios would be tested.

5.8.2 These were identified as Phase 1A which included a potential expansion of the supermarket store south of the A420 and a provision of employment land to the south. The second additional test, identified as Phase 2, assessed the full NEV development as per the Local Plan allocation.

Phase 1A

5.8.3 The results of the Phase 1A test are indicated below. This test assumes that an employment area of some 40ha to the south of the A420 is to be delivered together with an extension to the food superstore.

5.8.4 This scenario assumes an additional access onto the A420 and via a route through to Gablecross roundabout through the supermarket site.

5.8.5 The link flows are assessed at specific locations as indicated below. This comparison assumes that Phase 1A would be approved subsequent to Phase 1, such that Phase 1 becomes part of the baseline transport environment.

Table 5.11: Percentage Impact

Highway Link	2026 with Phase 1		2026 with Phase 1A		Percentage difference	
	AM Peak	PM Peak	AM Peak	PM Peak	AM Peak	PM Peak
M4 Motorway	7019	7746	7034	7692	0.2%	-0.7%
A419, J15 to Commonhead	4187	3817	4196	3800	0.2%	-0.4%
A4259 at Commonhead junction	3448	3248	3504	3136	1.6%	-3.4%
Purley Rd at Commonhead junction	601	492	591	444	-1.7%	-9.8%
Pack Hill at Commonhead junction	226	120	226	115	0.0%	-4.2%
A419, Commonhead to White Hart	5921	5404	5967	5412	0.8%	0.1%
A420 at White Hart	3680	3974	3923	4137	6.6%	4.1%

Ermin Street at White Hart	747	860	747	839	0.0%	-2.4%
Oxford Road at White Hart	1566	1580	1707	1602	9.0%	1.4%
Merlin Way at White Hart	2090	2329	2130	2402	1.9%	3.1%
A419 north of White Hart	6293	5785	6266	5698	-0.4%	-1.5%
Thornhill Road north of Gablecross	787	937	740	886	-6.0%	-5.4%
A420 east of Gablecross	2723	2662	2973	2754	9.2%	3.5%
A420 west of Gablecross	3508	3749	3831	3865	9.2%	3.1%
Supermarket/access S of Gablecross	584	957	689	929	18.0%	-2.9%
A420 east of Old Vicarage Lane	2251	2386	2157	2297	-4.2%	-3.7%
Old Vicarage Lane north A420	502	513	451	479	-10.2%	-6.6%
Old Vicarage Lane S of Highworth Rd	96	144	134	126	39.6%	-12.5%
Thornhill Road west of Highworth Rd	359	282	361	270	0.6%	-4.3%
Highworth Rd N of Old Vicarage Lane	376	343	383	315	1.9%	-8.2%
A420 west of New Eastern Access	2115	2229	1985	2206	-6.1%	-1.0%
A420 east of New Eastern Access	1982	2035	1909	2021	-3.7%	-0.7%
New Eastern Access Road	155	236	159	226	2.6%	-4.2%
Link between Honda RB and Hyde Rd	5593	5018	5575	5089	-0.3%	1.4%
Link between Hyde Road and Bypass	5504	4935	5479	4980	-0.5%	0.9%

5.8.6 The information contained above indicates the locations where there is predicted to be significant increases in traffic solely as a result of the addition of the Phase 1A development, i.e. that will exceed the IEMA thresholds of 10% increase along sensitive links. The following links meet that criteria:

- Supermarket/access south of Gablecross
- Old Vicarage Lane north A420
- Old Vicarage Lane south of Highworth Road

5.8.7 It is not unexpected that the flow along the Supermarket access increases as this corridor fundamentally forms part of the access strategy.

5.8.8 With regards to the impact along Old Vicarage Lane, a small change in flows will have a high impact where base flows are low, as is the case along Old Vicarage Lane.

Phase 2

5.8.9 The Phase 2 test reflects the delivery of the wider NEV development. In addition to Phase 1 and Phase 1A, Phase 2 includes an additional 6,000 dwellings, eight-form entry secondary school, primary schools and a GP surgery, dentist and pharmacy within local centres.

5.8.10 The modelling assumes that Phase 2 will be accessed via the following:

- A420, through the access delivered as part of Phase 1A
- Further access from A420 opposite Old Vicarage Lane
- An additional fourth arm onto the eastern access delivered as part of Phase 1
- Link to Commonhead roundabout

5.8.11 In addition to the access strategy identified above, various off site interventions have been including within the modelling. These include:

- Dualling of the A420 adjacent to the Police HQ
- Full signalisation of an enlarged Gablecross junction
- Diamond interchange at White Hart interchange
- A green bridge across the A419 to deliver a walking, cycling and public transport link
- Improvements to the Oxford Road/Drakes Way and Covingham Road/Dorcan Way transport corridors

5.8.12 The results of the Phase 2 test are indicated below.

Table 5.8: Percentage Impact

Highway Link	2026 with Phases 1 and IA		2026 with Phase 2		Percentage difference	
	AM Peak	PM Peak	AM Peak	PM Peak	AM Peak	PM Peak
M4 Motorway	7034	7692	6182	6556	-12.1%	-14.8%
A419, J15 to	4196	3800	3771	3721	-10.1%	-2.1%

Commonhead						
A4259 at Commonhead junction	3504	3136	3077	2918	-12.2%	-7.0%
Purley Rd at Commonhead junction	591	444	912	982	54.3%	121.2%
Pack Hill at Commonhead junction	226	115	330	1042	46.0%	806.1%
A419, Commonhead to White Hart	5967	5412	6138	5933	2.9%	9.6%
A420 at White Hart	3923	4137	1755	2606	-55.3%	-37.0%
Ermin Street at White Hart	747	839	685	699	-8.3%	-16.7%
Oxford Road at White Hart	1707	1602	981	1199	-42.5%	-25.2%
Merlin Way at White Hart	2130	2402	761	797	-64.3%	-66.8%
A419 north of White Hart	6266	5698	6477	6586	3.4%	15.6%
Thornhill Road north of Gablecross	740	886	1077	1150	45.5%	29.8%
A420 east of Gablecross	2973	2754	2178	1859	-26.7%	-32.5%
A420 west of Gablecross	3831	3865	3434	3512	-10.4%	-9.1%
Supermarket/access S of Gablecross	689	929	378	475	-45.1%	-48.9%
A420 east of Old Vicarage Lane	2157	2297	1541	1536	-28.6%	-33.1%
Old Vicarage Lane north A420	451	479	356	443	-21.1%	-7.5%
Old Vicarage Lane south of Highworth Rd	134	126	361	438	169.4%	247.6%
Thornhill Road west of Highworth Rd	361	270	511	545	41.6%	101.9%
Highworth Rd N of Old Vicarage Lane	383	315	382	405	-0.3%	28.6%
A420 west of New Eastern Access	1985	2206	1944	1788	-2.1%	-18.9%
A420 east of New Eastern Access	1909	2021	1943	1768	1.8%	-12.5%
New Eastern Access Road	159	226	195	197	22.6%	-12.8%
Link between Honda RB and Hyde Rd	5575	5089	5609	5226	0.6%	2.7%
Link between Hyde Road and Bypass	5479	4980	6033	4870	10.1%	-2.2%

5.8.13 The information contained above indicates the locations where there are predicted to be significant increases in traffic solely as a result of the addition of the Phase 2 development, i.e. that will exceed the IEMA thresholds of 10% increase along sensitive links. The following links meet that criteria:

- Purley Road at Commonhead junction
- Pack Hill at Commonhead junction
- A419 North of White Hart
- Thornhill Road north of Gablecross
- Old Vicarage Lane south of Highworth Road
- Thornhill Road west of Highworth Road
- Highworth Road north of Old Vicarage Lane
- New eastern access road
- Link between Hyde Road and Bypass

5.8.14 The results of the phase 2 modelling indicate that there is a significant increase in traffic along the Purley Road and Pack Hill corridors adjacent to the Commonhead junction. This is not unexpected as the access strategy for Phase 2 includes a link to this junction and would therefore invariably increase traffic at this location. This increase has the potential to affect the criteria identified within the IEMA Guidance.

5.8.15 However, it is envisaged that these corridors, being a point of access into the Phase 2 area, will be subject to highway improvements. This will include improving the provision for all modes of travel, including walking and cycling. This in all likelihood will include the provision of adequate walking and cycling links, safe crossing points and highway improvements to ensure acceptable highway link capacity. With appropriate mitigation, which will be defined at the appropriate planning stage, it is considered that there will not be any significant residual adverse impact at these locations.

5.8.16 High percentage increases in traffic flow are also predicted in relation to the links around Highworth Road. The high increases are due to the relatively low level of background traffic and the fact that any small increase in flow would result in a high percentage increase. The levels of traffic flow that are predicted once phase 2 is included are within the highway capacity of the associated roads. Furthermore, additional mitigation measures could be considered at the appropriate time to influence route choice to avoid South Marston. It is considered that there will not be any significant adverse residual effect once such appropriate mitigation has been considered.

5.8.17 In relation to the other routes that are identified, the predicted increases are close to the threshold and as such are unlikely to have a significant impact. Furthermore, various interventions are identified within the Local Plan, including corridor improvements and the Green Bridge, all of which will mitigate the impact of phase 2 together with any additional mitigation that is defined during detailed modelling stages.

Phase 1A and 2 Summaries

5.8.18 Both Phase 1A and Phase 2 have been modelled. These are preliminary results as further detailed modelling would be carried out to support any future planning applications for those developments. The results of the modelling have not highlighted any impact that cannot be appropriately mitigated. Furthermore, the delivery of the Proposed Development as Phase 1 of the NEV will not prejudice the delivery of any subsequent phases.